



WordClock 16x16 - 2024



AWSW

[VIEW IN BROWSER](#)

updated 10. 9. 2026 | published 10. 9. 2026

Summary

Solder free, new case with less parts, USB-C, new 2nd font, new ESP32, faster setup, better higher intensity usage

[Hobby & Makers](#) > [Electronics](#)

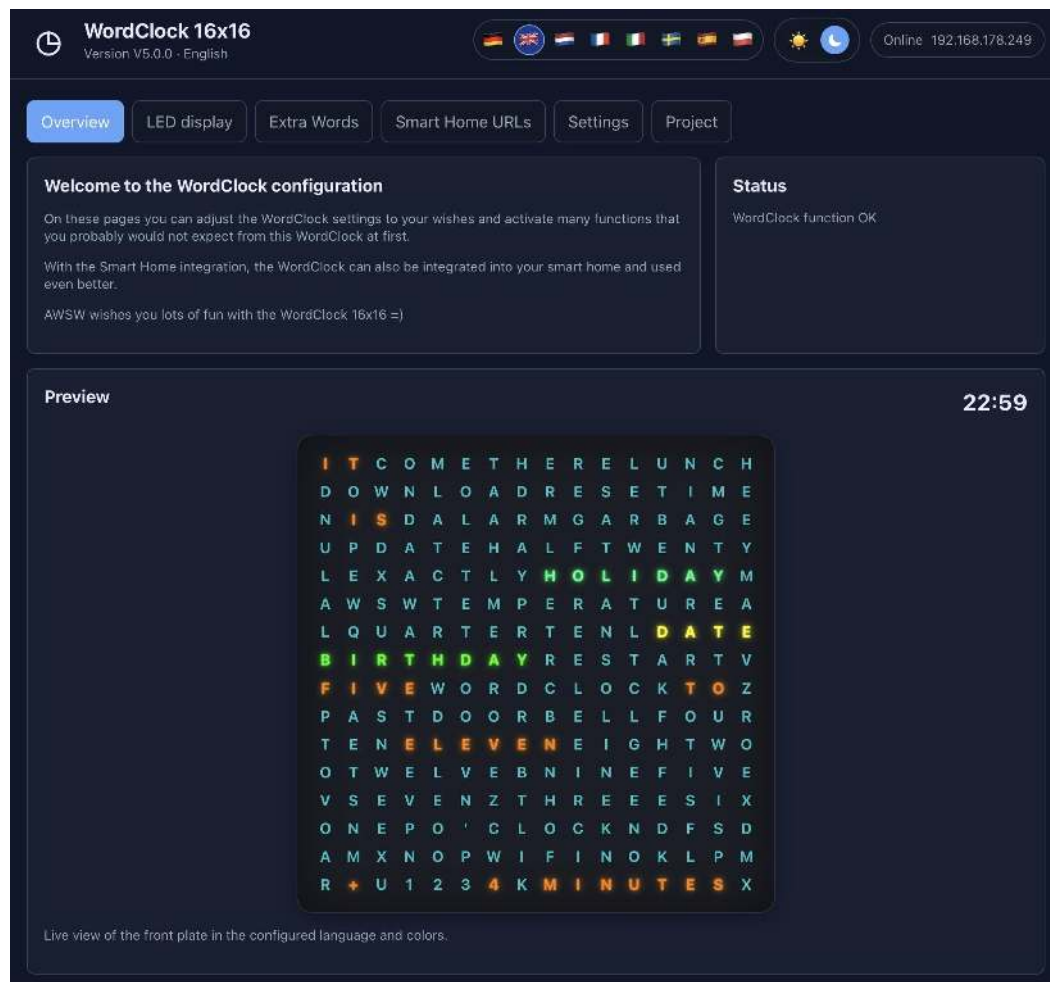
Tags: [clock](#) [time](#) [arduino](#) [esp32](#) [wordclock](#) [16x16](#)
[wortuhr](#) [16x16ledmatrix](#) [aws](#)

WordClock 16x16 - 2024

This is the 2024 variant to the [WordClock 16x16](#) model adding new functions and features as well as a new font as V2 variant. =)

This WordClock 2024 variant covers mostly every input I got from users in the last 2 years for the previous model. =)

BIG news on 11.07.2026: Software version V5.0.0 available!



WordClock 16X16 received its biggest update yet. I fully rewrote the configuration portal and added lots of new features and improvements to the software.

To name a few of them please read this list and have a look at the pictures below.

I am really looking forward to know your feedback and impressions.

As you might expect this was a tone of work and you can download the new update for free via the internal software update of your WordClock or with the ESP32 flash script as usual.

Enjoy and I hope you will like it 😊👍😊

WordClock 16x16 V5.0.0 new functions:

Complete redesign of the web interface with much more modern design with lots of new features:

- New Welcome page with Status section on the start page to see if there are configuration problems or new software updates available.
- Preview function of the WordClock front plate with set colors and live show time and Extra Word texts.
- Responsive design with dynamically adjusted sections and content.
- Multi language support fitting to the 8 available languages of WordClock 16x16. Just use your WordClock in your desired language now. All texts were translated in the web configuration to your local language.
- Day and Night mode of the web configuration in a light or dark design.
- The Extra Words can be set now on and off together with their color settings directly in the web interface.
- Complete redesigning Smart Home URLs page including test and URL copy button functions with tons of new options replacing the old text based list.
- Complete redesigned page when entering the `http://WordClock-IP-Address:2023` page.
- Fully redesigned notifications for activation and deactivation of the classic Offline Mode with much more useful hints about this functionality.
- New Profile page with some more useful information to the WordClock 16x16 project and my other projects on Printables or GitHub.

Other new functions:

- New setup method via WiFi WPS function before the known WiFi Captive Portal is set up. Just click the WPS connect button on your router or access point to add your WordClock to your network. After this you can set the language and other settings within the new web portal. WordClock will guide you through the process and tells you the received IP-address on startup to complete the setup directly in the new web configuration portal. No WiFi password entering needed any more. =)
- New scrolling text function which was much requested to add dynamic texts to the display in BIG letters. This is possible to setup, test in the web interface and the resulting

URL can be copied with 1 click to add to your own smart home environment as http-URL as with the classic versions of WordClock.

- Random color time texts and background color if no Extra Words are currently shown and the device is in Day Mode.
- Lots of smaller bug and problem fixes, e.g. if the time server is no longer available. The configuration corrects itself automatically now.
- Improved reconnecting of the portal to the device after restarts.
- Separate colors for Day and Night mode for the time text and the background color.
- Separate brightness settings for the time text and the background LEDs allowing to use the same color for both with different brightness settings to create totally new effects and impressions.
- Optional digital time output as e.g. 14:30 in big letters on the whole LED matrix on half and full hours.
- Optional automatically performed searches for new software updates with notifications on the LED display and in the web interface.
- New software update animation to show the progress of the update process.
- New startup texts and informations on the LED display with bigger and better readable texts.
- New maximum amount settings for WiFi connection and time server tries before the configuration is automatically reset to be able to use the WordClock again without flashing it in case you switched your WiFi router or the time server is no longer available.
- Smooth LED word transitions to get a more refined look of the time piece.
- The software update is compatible with all WordClock 16x16 variants from me including the initial one from 2023, over the most famous 2024, up to the latest 2025 variant. So the good old "D1 mini ESP32" can run the same software as the from 2024 and up to now used "NodeMCU ESP32" with no limited functionality. To my experience from weeks of testing and using the new software version both will run better than ever before. =)

Removed functions:

- The „Reset WiFi only“-function was removed because it was integrated into the complete new „Reset all settings“-function for improved usage.

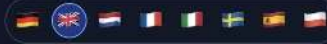
- Telegram support needed to be removed to be able to get more memory in the ESP32 to add the new web portal and to free up more RAM inside the ESP32 to be able to use all these new functions with much more performance. Because this was always more a gimmick than a useful function I think not many will miss it...

Here are some pictures from software V5.0.0:



WordClock 16x16

Version V5.0.0 - English



Online 192.168.178.249

Overview

LED display

Extra Words

Smart Home URLs

Settings

Project

Welcome to the WordClock configuration

On these pages you can adjust the WordClock settings to your wishes and activate many functions that you probably would not expect from this WordClock at first.

With the Smart Home integration, the WordClock can also be integrated into your smart home and used even better.

AWSW wishes you lots of fun with the WordClock 16x16 =>

Status

WordClock function OK

Preview

22:59

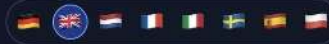
I T C O M E T H E R E L U N C H
D O W N L O A D R E S E T I M E
N I S D A L A R M G A R B A G E
U P D A T E H A L F T W E N T Y
L E X A C T L Y H O L I D A Y M
A W S W T E M P E R A T U R E A
L Q U A R T E R T E N L D A T E
B I R T H D A Y R E S T A R T V
F I V E W O R D C L O C K T O Z
P A S T D O O R B E L L F O U R
T E N E L E V E N E I G H T W O
O T W E L V E B N I N E F I V E
V S E V E N Z T H R E E S I X
O N E P O ' C L O C K N D F S D
A M X N O P W I F I N O K L P M
R + U 1 2 3 4 K M I N U T E S X

Live view of the front plate in the configured language and colors.



WordClock 16x16

Version V5.0.0 - English



Online 192.168.178.249

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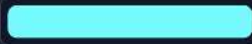
Colors

Day

Time



Background

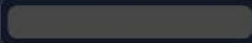


Night

Time



Background



The night colors are shown at night when night mode is on.

Show single minutes



Random day colors



When no extra words are active, the time and background colors are picked randomly every minute in day mode, with good contrast so the time stays readable.

Smooth LED word transition



When words and minutes change, the LEDs crossfade smoothly instead of switching hard.

Day/night brightness

Use night mode

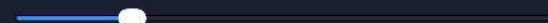


Day

Time: 25



Background: 10

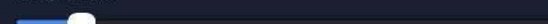


Night

Time: 25



Background: 5



Day starts

Day ends

06:15

22:14

Current: Night time

Digital time

Show digitally on the full and half hour



Shows the time on the matrix for 15 seconds as a digital display on the full and half hour.

Test digital time

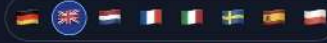
Layout language

Layout language

Englisch



The language must match the letter layout of the front plate.



Overview

LED display

Extra Words

Smart Home URLs

Settings

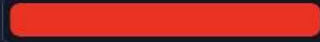
Project

Extra Words

Activation and colors for all words of the selected layout.

1: COME HERE

Color



Active



2: LUNCH TIME

Color



Active



3: ALARM

Color



Active



4: GARBAGE

Color



Active



5: HOLIDAY

Color



Active



6: TEMPERATURE

Color



Active



7: DATE

Color



Active



8: BIRTHDAY

Color



Active



9: DOORBELL

Color



Active





WordClock 16x16

Version V5.0.0 - English



Online 192.168.178.249

Overview

LED display

Extra Words

Smart Home URLs

Settings

Project

Show text

Enter a line of text — the WordClock shows it as scrolling text in the chosen color and brightness. Umlauts and special characters (e.g. äöüß !\$%?) are supported. Copy one of the two URLs (with the clock's current IP) into your smart-home system.

Text to display

This is the text to display

Text color



Test

Scrolling text brightness: 40



URL (with quotes)

http://192.168.178.249:2023?text="This is the text to display"

Copy

URL (percent-encoded)

http://192.168.178.249:2023?text=This%20is%20the%20text%20to%20display

Copy

Smart Home URLs

Control Extra Words, colors and brightness through the smart-home URLs.

The embedded smart-home URL page uses the same language as this interface.

Extra Words

Quick controls for extra words, colors and brightness. Existing smart-home URLs stay unchanged.

Live Status

☒ Time text #0056D6

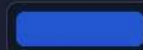
☐ Background #333333

Brightness: 25

API status: </status>

Colors and brightness

Time text



Background



Brightness



25

Apply now

This sets brightness as web override.

Shortcuts

Hide all until
midnight

Enable all
again

Reset status

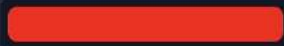
Device: esp32-6A598D at 192.168.178.249

Extra Words

1: COME HERE

OFF

Color



Current color #FF0000

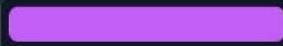
On

Off

2: LUNCH TIME

OFF

Color



Current color #D357FE

On

Off

3: ALARM

OFF

Color



Current color #FFFF00

On

Off

4: GARBAGE

OFF

Color



Current color #0000FF

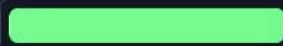
On

Off

5: HOLIDAY

ON

Color



Current color #00FF00

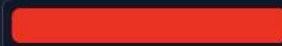
On

Off

6: TEMPERATURE

OFF

Color



Current color #FF0000

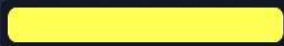
On

Off

7: DATE

ON

Color



Current color #FFFF00

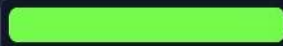
On

Off

8: BIRTHDAY

ON

Color



Current color #00FF00

On

Off

9: DOORBELL

OFF

Color



Current color #0000FF

On

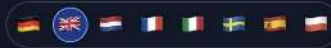
Off

Smart-home URLs

The existing automation endpoints continue to work unchanged.

Ticker text: The parameter `?text=This%20is%20the%20text` shows a dynamic text in the current time-text color. Quotes such as `?text="Hello"` are allowed.

Description	URL		
General			
Get status	<code>/status</code>	Test	Copy
Brightness mode	<code>/intensity</code>	Test	Copy
Show scrolling text	<code>?text=This%20is%20the%20text%20to%20display</code>	Test	Copy
Colors and brightness			
Time text blue	<code>/config?R-Time=0&G-Time=0&B-Time=255</code>	Test	Copy
Background red	<code>/config?R-Back=255&G-Back=0&B-Back=0&INTENSITY=5&INTENSITYviaWEB=1</code>	Test	Copy
Display off	<code>/config?INTENSITY=0&INTENSITYviaWEB=1</code>	Test	Copy
Back to auto mode	<code>/config?INTENSITY=25&INTENSITYviaWEB=0</code>	Test	Copy
All extra words			
All on	<code>/ew/?ew1=1&ew2=1&ew3=1&ew4=1&ew5=1&ew6=1&ew7=1&ew8=1&ew9=1&ew10=1&ew11=1&ew12=1</code>	Test	Copy
All off	<code>/ew/?ew1=0&ew2=0&ew3=0&ew4=0&ew5=0&ew6=0&ew7=0&ew8=0&ew9=0&ew10=0&ew11=0&ew12=0</code>	Test	Copy
All off until midnight	<code>/resetew1</code>	Test	Copy
Reactivate all	<code>/resetew0</code>	Test	Copy
Reset status	<code>/resetewstatus</code>	Test	Copy
Single extra words			
EW on 1	<code>/ew/?ew1=1</code>	Test	Copy
EW off 1	<code>/ew/?ew1=0</code>	Test	Copy



Overview

LED display

Extra Words

Smart Home URLs

Settings

Project

Maintenance

Restart

Restarts the ESP32. Saved settings are preserved.

Reset all settings

Resets the WordClock configuration to defaults and starts setup again.

Startup display

Show WordClock start text



Show IP address at startup



Software update

Installed

V5.0.0

Available

V5.0.0

Status

You are using the latest available software.

Check for update

Start update now

Automatic update check



Not automatic



5 minutes after restart



Daily on the hour

If an update is found, the letters WORDCLOCK light up cyan on the clock as a reminder.

Hour

22



Minute

00



Time server

NTP server

Default: pool.ntp.org



A change restarts the WordClock.

Time zone

POSIX time zone

Europe: Berlin / Paris / Rome



Active: CET-1CEST,M3.5.0,M10.5.0/3

A change restarts the WordClock.

Wi-Fi

SSID	FRITZ!Box 7590 KE
Signal	-79 dBm
Hostname	esp32-6A598D
MAC	C4:4F:33:6A:59:8D
Gateway	192.168.178.1
DNS	192.168.178.5

Maximum connection attempts when starting the WordClock

Maximum Wi-Fi connection attempts

10



After this number is reached, the Wi-Fi settings are deleted. Initial setup must then be performed again because the Wi-Fi router is apparently no longer available.

Maximum time server connection attempts

10



After this number is reached, pool.ntp.org is automatically saved as the time server.

Other functions

Test mode

Starts the LED test mode to check the matrix and colors directly on the WordClock.

Reset all Extra Words

Switches all Extra Words back without changing other settings.

Operating mode

Here you choose whether the WordClock uses your normal Wi-Fi or starts its own offline access point.

Use online mode

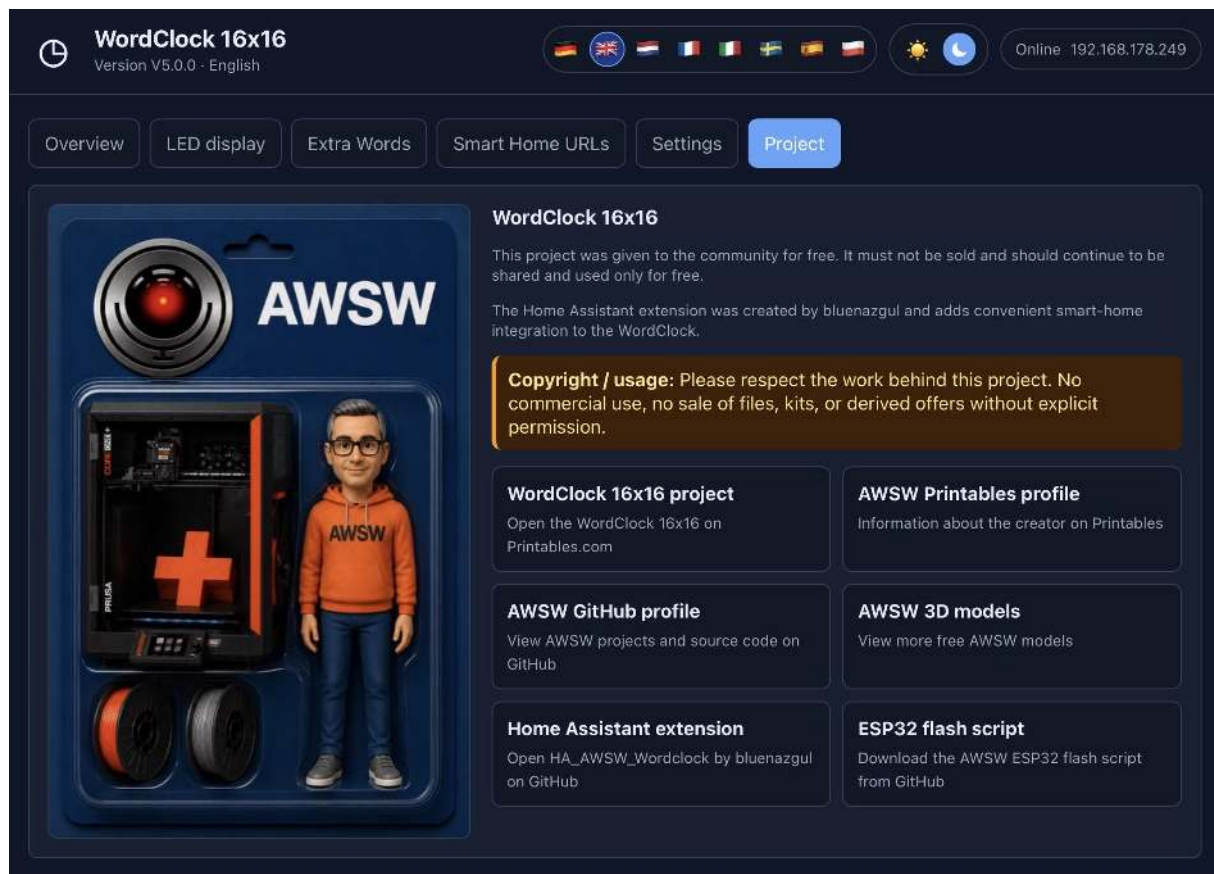
The WordClock is currently in online mode.



Online: Connection to the saved Wi-Fi, reachable through router IP or hostname.

Offline: own access point **WordClock in Offline Mode**, Password **Matrix-16x16**, Web address **http://192.168.4.1**.

A change restarts the WordClock and is applied only after confirmation in the dialog.



Here is the rest of the previous article focusing on notes to the software versions previous to V5.0.0:

New functions and options:

- New solder free build option → Yes, you do not need to solder if you like not to do so!
- New design with wider, thicker case as shown in the pictures → more stable on desks.
- New sliding case that can very easily be opened again if needed.
- USB-C port for power input. This ensures more stable power with the better USB-A to USB-C cables available to avoid many of your power issues. Who remembers the “Brownout detector” message in the Arduino IDE or continuously rebooting ESP32 boards does know the reason for this change. ;-)
- New more round font front named as “V2” as a new option.
- New ESP32 usage with USB-C with better power options, also to avoid many issues.
- Faster setup through less actions, easy wiring with less tools.
- Better higher intensity usage, because the LED Matrix is directly powered now. No poor dying ESP32 boards anymore due to too bright colors and/or to high intensity values...

- Easy assembly like counting 1,2,3 with detailed descriptions below. It seems more complicated at first, but you will get it easily done or can't you count 1,2,3? ;-)
- Many hints inside the case to help you with the setup.
- FAQ section to most asked questions listed below.

All features and the code are the same as for the **WordClock 16x16** model. So “just” the design is changed and needed quite a few changes to make this work. It again uses a printed diffuser instead of the piece of paper like the previous model. This model requires 2 color changes during printing of the front part. More to that later...

These languages are available in the font variants V1/V2:

1. German
2. English
3. Dutch
4. French
5. Italian
6. Swedish
7. Spanish
8. Polish (2024 models only)

Note: New upcoming languages might be added to the 2024 model only.

Hope you like this new variant! =)

Notes:

- **20.07.2026** - New way to install the software:
 - Method 1 (NEW):
 - Use the ESP32 Web Flasher: <https://aws-projects.de/esp32-web-flasher/>
 - Method 2:
 - Please use the **ESP32 flash script** to update your device.
- **10.11.2024:** Added a **new script to flash the ESP32** without the need of the Arduino IDE:

<https://www.youtube.com/watch?v=oiCbp8OFSus>

- **21.12.2024: Software update V4.2.0** is now available including several new functions:
 - **Direct software update** without the need of an computer or the Arduino IDE as introduced with **WordCalendar 2024** before. Watch **this video** for details. You will need to select the used ESP32 board after the update to this version to download the

correct files for your used ESP32 in future. A new page asking for this information will be shown when you enter the web configuration portal after the update.

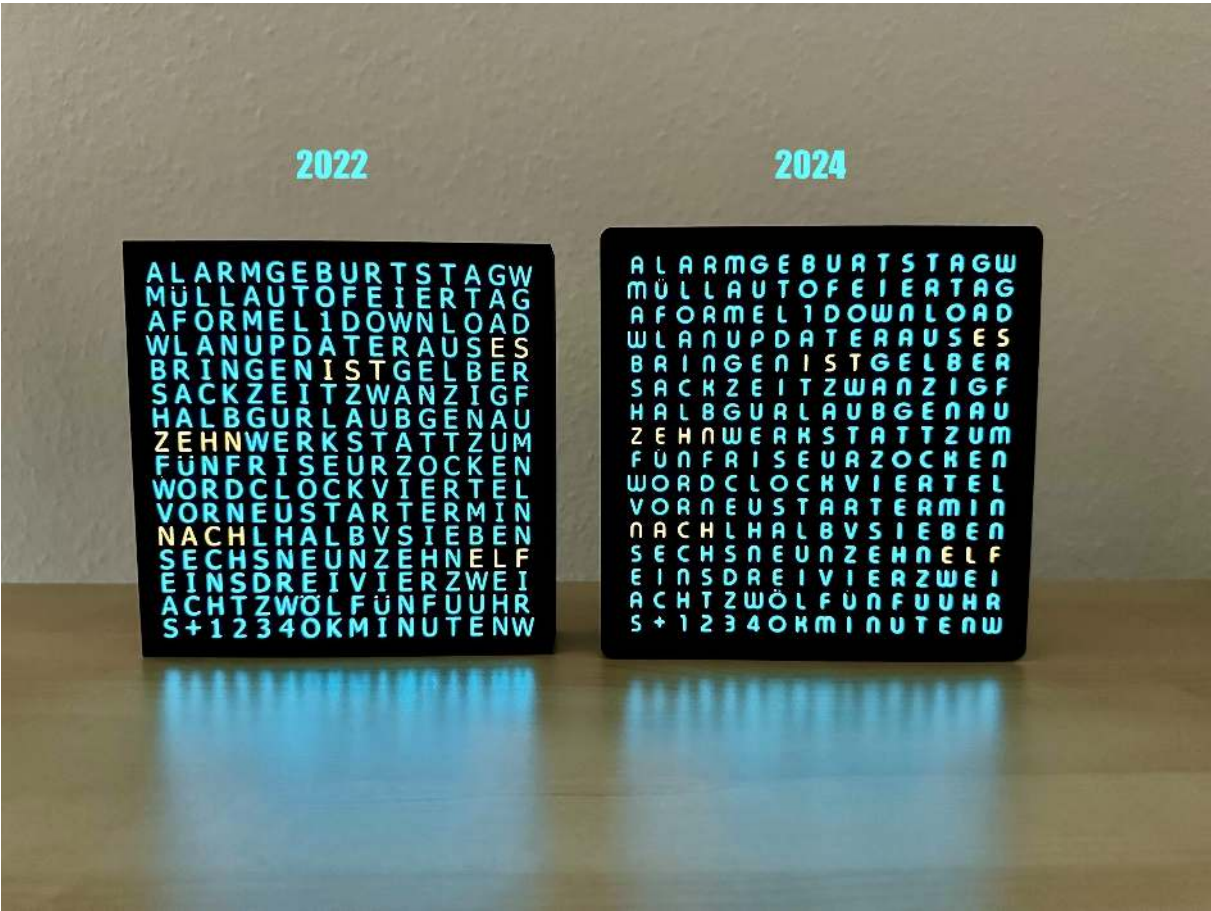
- **Scrolling texts function** that shows dynamic texts like the IP-address or the initial startup messages much better readable. Watch [this video](#) for details.
- New **startup and initial setup texts** included.
 - The installed software version is now shown during startup in case the “WordClock” text is set active.
 - The IP-address information to enter the web configuration portal should be shown more clear to read now.
- **Faster startup** on reboot.
- Link to an **Telegram HowTo** added to let you know how you can add this function.
- Added a “ **test mode**” in the “Maintenance” section to check all time word output texts faster.
- Lots of **small changes to the web configuration portal** to make the device better.
- To update to this new version from older software versions you will need to use the [ESP32 flash script](#) once to get the new software update variant. If you just new installed your WordClock you will get the newest version with the script already. =)
- **05.01.2025:** You will now find the ESP32 flash script as a variant of the known MS Windows script for the usage in macOS too. Thanks a lot to **John (Rave Props)** @JohninNL_289006 for adding this macOS variant! This should make it even more simple to setup the project for macOS users. =)
- **19.06.2025:** New layout for the 2024 models available to support the Polish language with software version V4.8.0. This language is not supporting the WordClock 16x16 of 2025 font unfortunately, because the 2025 font does not support the needed Polish letters fully:

RESETEMPERATURE
RWOLNEUPDATEJEST
AURODZINYŚMIECIX
WAKACJEALARMWPÓŁ
SDATAXCZASOBIAD
DWADZIEŚCIATWIFI
KWADRANSZWONEKV
PIĘĆDZIESIĘĆCAWSW
POBIERZWORDCLOCK
DOKŁADNIEPODEJDŹ
DRUGAPIĄTASZOSTA
SIÓDMADZIESIĄTAN
JEDENASTACZWARTA
DZIEWIĄTATRZECIA
PIERWSZADWUNASTA
ÓSMAP+U1234TMINJ

RESETEMPERATURE
RWOLNEUPDATEJEST
AURODZINYŚMIECIX
WAKACJEALARMWPÓŁ
SDATAXCZASOBIAD
DWADZIEŚCIATWIFI
KWADRANSZWONEKV
PIĘĆDZIESIĘĆCAWSW
POBIERZWORDCLOCK
DOKŁADNIEPODEJDŹ
DRUGAPIĄTASZOSTA
SIÓDMADZIESIĄTAN
JEDENASTACZWARTA
DZIEWIĄTATRZECIA
PIERWSZADWUNASTA
ÓSMAP+U1234TMINJ

Here are some pictures mostly in German and some parts in light color for a better view:

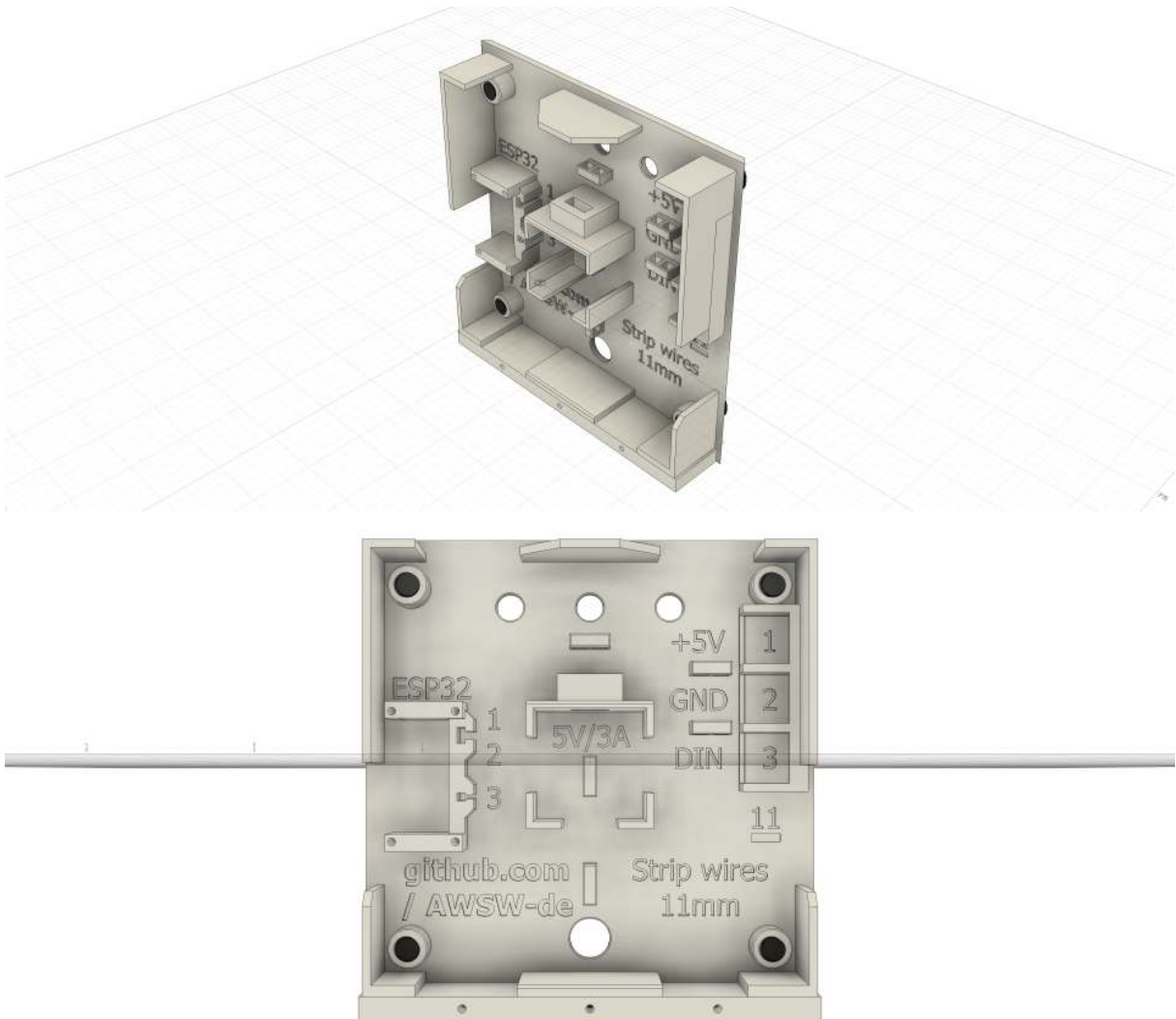




ITCOMETHEREELUNCH
DOWNLOADRESETIME
NISDALARMGARBAGE
UPDATEHALFTWENTY
LEXACTLYHOLIDAYM
AWSWTEMPERATUREA
LQUARTERTENLDATE
BIRTHDAYRESTARTV
FIVEWORDCLOCKTOZ
PASTDOORBELLFOUR
TENELEVENEIGHTWO
OTWELVEBNINEFIVE
VSEVENZTHREEESIX
ONEPO'CLOCKNDFS
AMXNOPWIFINOKLPM
R+U1234KMINUTESX

IT COME THERE LUNCH
DOWNLOAD RESET TIME
NISDALARMGARBAE
UPDATEHALFTWENTY
LEXACTLYHOLIDAYM
AWSWTEMPERATUREA
LQUARTERTENLDATE
BIRTHDAYRESTARTV
FIVEWORDCLOCKTOZ
PASTDOORBELLFOUR
TENELEVENEIGHTWO
OTWELVEBINEFIVE
VSEVENZTHREESIX
ONEPO'CLOCKNDFS
AMXNOPWIFINOKLPM
R+U1234KMINUTESX





As shown the inside of the lid now contains several new elements to fasten the new ESP32 board, reducing the amount of parts drastically and allowing to easily setup your WordClock without any soldering actions. More on that below...

Functions:

- WordClock using a 16x16 LED matrix and regular fonts.
- Super easy print and build, 3 wires to solder or attach solder free in this new version only.
- Video instructions available to help you with the code and WiFi setup.
- Small print size 180x180 to fit well on the Prusa MK3S+ or any 200x200mm printer. It should fit on the Prusa MINI good as well.
- The above named language variants are available using the same code for all of them.
- The case can be placed on a desk without a stand or on your wall.
- The case is provided in 2 main parts in general to hold the 16x16 matrix and the ESP32 board inside.

- The 4 optional printed screws generate a gap between the wall and the case to get the power cable out straight and to get air in the case from the back.
- The diffuser layer to show the LED colors is printed in white PLA with 2 color changes of the front part.
- No supports needed to print all parts. The parts just need to be placed on their flat sides.
- The lid part slides in place into the front part and presses the LED matrix into the holes for each letter.
- Web configuration interface to control the WordClock functions.
- Custom created WiFi manager integration to avoid setting your WiFi credentials manually in the code.
- The from your WiFi router received IP-address is shown as text on the display to inform you about the current value.
- During startup a text for "WIFI" or "WLAN" is shown in cyan color until the WiFi settings were done for the first time. Afterwards blue until the device received a new IP-address then in green color.
- In case the configured time server cannot be reached a red "TIME" text will be shown and the device will restart until it can reach the time server.
- The configuration can be set with an integrated web portal with the ESP32 own host name like "http://esp32-D88998" or via IP-address like "http://192.168.178.50" (depending on your ESP32 hostname and your router settings).
- Operation Mode: "Online Mode" with WiFi usage or in "Offline Mode" without the need of a local WiFi possible. In both cases the internal web configuration portal is available.
- The LED color for the time texts and the background can be configured to a static value.
- Also available is an option to set a random text color every new minute.
- The LED intensity can be set for a day time and optional for a night mode usage. With that a time span can be selected to turn off the LEDs fully or reduce their intensity for night time usage.
- Updates of the software can be done "Over The Air" / "OTA" with the internal web portal as well to avoid connecting the device to Arduino IDE again. See instructions below.
- The in the internal Wifi manager stored settings can be set to default on request to switch to a new WiFi router afterwards.
- The WordClock settings can be set to default as well. WiFi and some other basic settings will be kept.
- In case the WiFi is no longer available the device will reset the WiFi settings automatically after 30 seconds to avoid flashing with Arduino IDE again. Follow the initial WiFi setup again.
- Smart Home environment integration to turn the LED display OFF and ON with HTTP url commands to save some power when e.g. you are

not at home. The HTTP commands can be viewed from the web configuration portal.

- Telegram functionality to control the extra texts and notify you about events.
- Smart home integration: Several additional texts in different colours can be displayed together with the time to notify you about events:

DE:	EN:	Other languages:
Geburtstag	Birthday	x
Alarm	Alarm	x
Feiertag	Holiday	x
Urlaub	Holiday	x
Müll raus bringen	Garbage day	x
Termin	Date	x
Formel1	-	-
Auto	-	-
Zeit zum Zocken	-	-
Gelber Sack	-	-
Werkstatt	-	-
Friseur	-	-
-	Lunch time	x

-	Come here	x
-	Doorbell	x
-	Temperature	x

Integrated web configuration interface:

This WordClock can be used in "Online Mode" with WiFi usage or in "Offline Mode" without the need of a local WiFi. In both modes you will be able to use an internal configuration portal to configure your WordClock. See pictures below. =)

Online Mode:

WordClock **Connected**

General:

WordClock configuration

Welcome to the WordClock configuration. Here you can adjust your WordClock settings to your personal needs. Enjoy your WordClock =)

WordClock software version

V3.2.0

Operation mode:

Operation mode

WordClock is used in Online Mode. All functions are available.

Use WordClock in Online Mode (change forces restart)



Show the Operation Mode hints

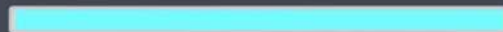


LED settings:

Time



Background



Show single minutes to display the minute exact time



Day/Night LED brightness mode settings:

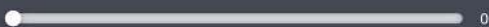
Use night mode to reduce brightness



Brightness during the day



Brightness at night



Night mode status

Day time

Day time starts at

06:15

Day time ends after

21:14

Startup:

Show the 'WordClock' text on startup



Show IP-address on startup

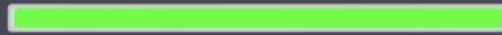


Extra Words:

1: ALARM



2: GEBURTSTAG



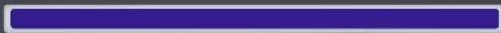
3: MÜLL RAUS BRINGEN



4: AUTO



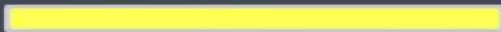
5: FEIERTAG



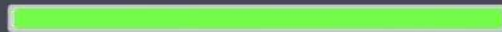
6: FORMEL1



7: GELBER SACK



8: URLAUB



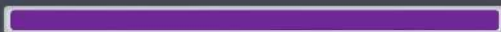
9: WERKSTATT



10: ZEIT ZUM ZOCKEN



11: FRISEUR



12: TERMIN



WiFi:

SSID FRITZ!Box 7590 KE	Signal -53dBm
Hostname in your router esp32-D88998	MAC address F0:08:D1:D8:89:98
IP-address 192.168.178.9	DNS address 192.168.178.1
Gateway address 192.168.178.1	

Time settings:

WordClock startup time Friday, February 16 2024 11:58:24	Choose your time server (change forces restart) Best choice: Your local router (WiFi gateway address) ⌵
Used NTP time server 192.168.178.1	Choose your time zone (change forces restart) Central Europe: CET-1CEST,M3.5.0,M10.5.0/3 (Austria,Denmark,Franc ⌵
Used time zone value CET-1CEST,M3.5.0,M10.5.0/3	

Telegram support:

Use Telegram support (change forces restart) ☒	React to your own Telegram ID only ☒
Send notifications about messages from unknown users ☒	Show your own Telegram bot token and chat id ☐

Smart home control via web URLs:

About note

Control WordClock from your smart home environment via web URLs.

Functions

You can configure colors, the LED intensity and the extra words and get the status of each named element.

Usage hints and examples

<http://192.168.178.9:2023> or <http://esp32-D88998:2023>

Update:

Activate update mode

Activate update mode

Update URLs

<http://192.168.178.9:8080> or <http://esp32-D88998:8080>

Download newer software updates here

<https://github.com/AWSW-de/WordClock-16x16-LED-matrix>

Language:

Change layout language

German

Current layout language

German

Maintenance:

Restart WordClock

Restart WordClock

Reset settings to enter initial values again

Reset WiFi, language, Telegram settings

Reset all Extra Words

Reset all Extra Words

Reset settings (except WiFi, language, Telegram & operation mode)

Reset WordClock settings

License information:

License information

NonCommercial – You may not use the project for commercial purposes!

Offline Mode:

WordClock **Connected**

General:

WordClock configuration

Welcome to the WordClock configuration. Here you can adjust your WordClock settings to your personal needs. Enjoy your WordClock =)

WordClock software version

V3.2.0

Operation mode:

Operation mode

WordClock is used in Offline Mode. Please check time value.

Use WordClock in Online Mode (change forces restart)



Get time from your device

Get time from your computer, phone, tablet

Time received from your device

8:7:52

Current hour offset

+ 1 hours



Time set on WordClock

9:7:52

Show the Operation Mode hints

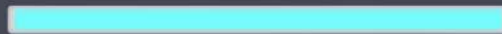


LED settings:

Time



Background



Show single minutes to display the minute exact time



Day/Night LED brightness mode settings:

Use night mode to reduce brightness ☒	Brightness during the day <input type="range" value="15"/>
Brightness at night <input type="range" value="0"/>	Night mode status Day time
Day time starts at 06:15	Day time ends after 21:14

Language:

Change layout language German	Current layout language German
---	--

Maintenance:

Restart WordClock Restart WordClock	Reset all Extra Words Reset all Extra Words
Reset settings (except WiFi, language, Telegram & operation mode) Reset WordClock settings	

License information:

License information
NonCommercial — You may not use the project for commercial purposes!

Setup the code for the WordClock and WiFi setup:

- Follow this video to get used to the usage:

<https://www.youtube.com/watch?v=oiCbp8OFSus>

```
Administrator: Windows PowerShell

#####
# # # ##### # # ##### ##### #####
# # # # # # # # # # # # # # #
# # # # # # # # # # # # # # #
##### # # # # # # # # # # # # #
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# # # # # # # # # # # # # # #
##### # # # # # # # # # # #

AWSW ESP32 Flash Script - V1.1.0

This script can flash your ESP32 board fully automatically without the need of installing Arduino IDE or any library.
Everything will be downloaded and executed for you... You just need to wait some minutes until the process is finished.
! IMPORTANT: You are using this script at your own risk. The manual way to do this is well described too !

STEP 1 OF 8 - Select your project to be flashed to the ESP32:

Choose your AWSw project to flash to the ESP32:

1. WordClock 16x8 - 2023 (ESP32 D1 mini)
2. WordClock 16x16 - 2023 (ESP32 D1 mini)
3. WordClock 16x8 - 2024 (ESP32 NodeMCU)
4. WordClock 16x16 - 2024 (ESP32 NodeMCU)
5. WordCalendar - 2024 (ESP32 NodeMCU)

E - Erase all ESP32 flash content only
X - Exit the script without flashing the ESP32

Choose the project to flash in the next steps to the ESP32:
```

Code download:

- The script to upload the code can be downloaded from my [GitHub repository](#).
- See the **FAQ** section too for more details to the code software.

Update the code of a running WordClock:

- Updates can be done “Over The Air” / “OTA” with the internal WordClock web portal.
- You will find a section to get the update without the need of a computer with 2 clicks.

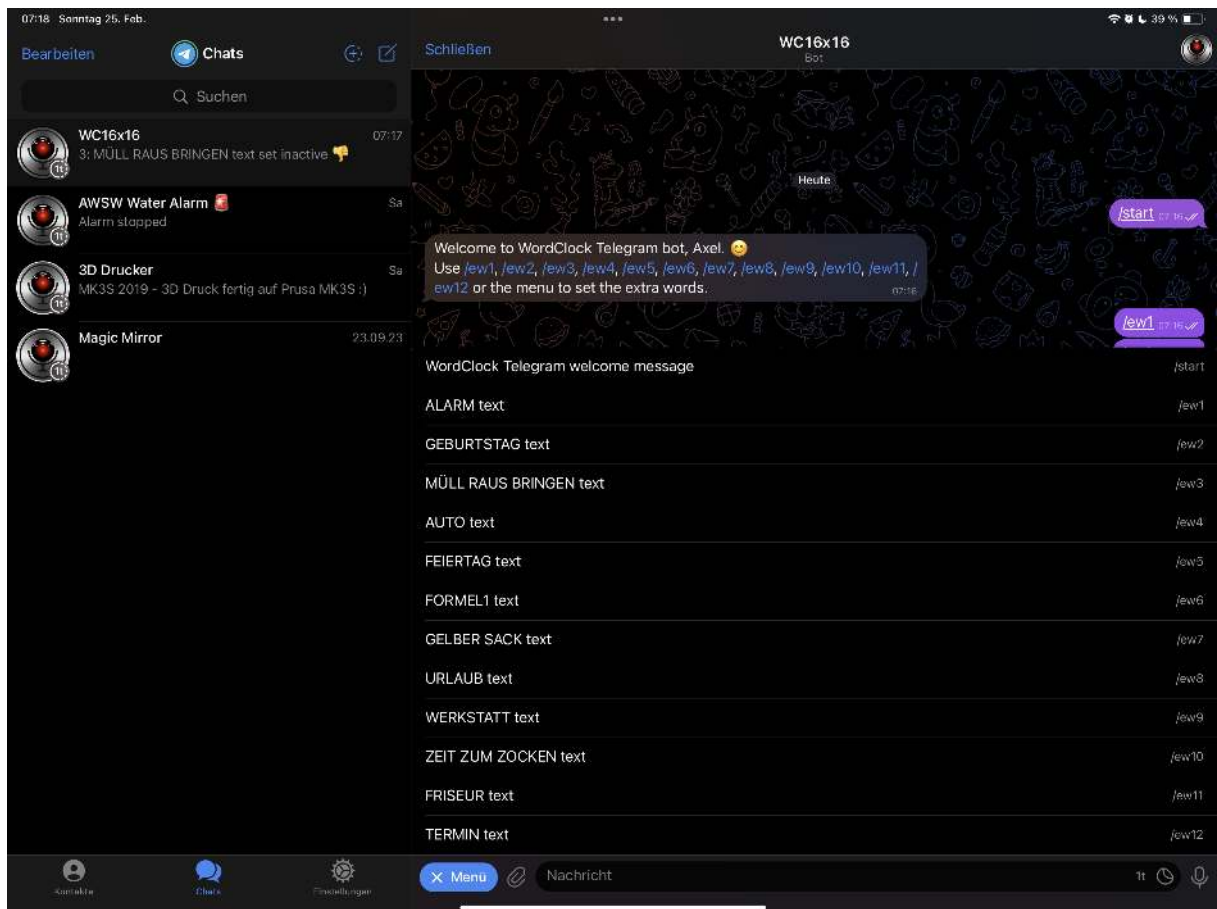
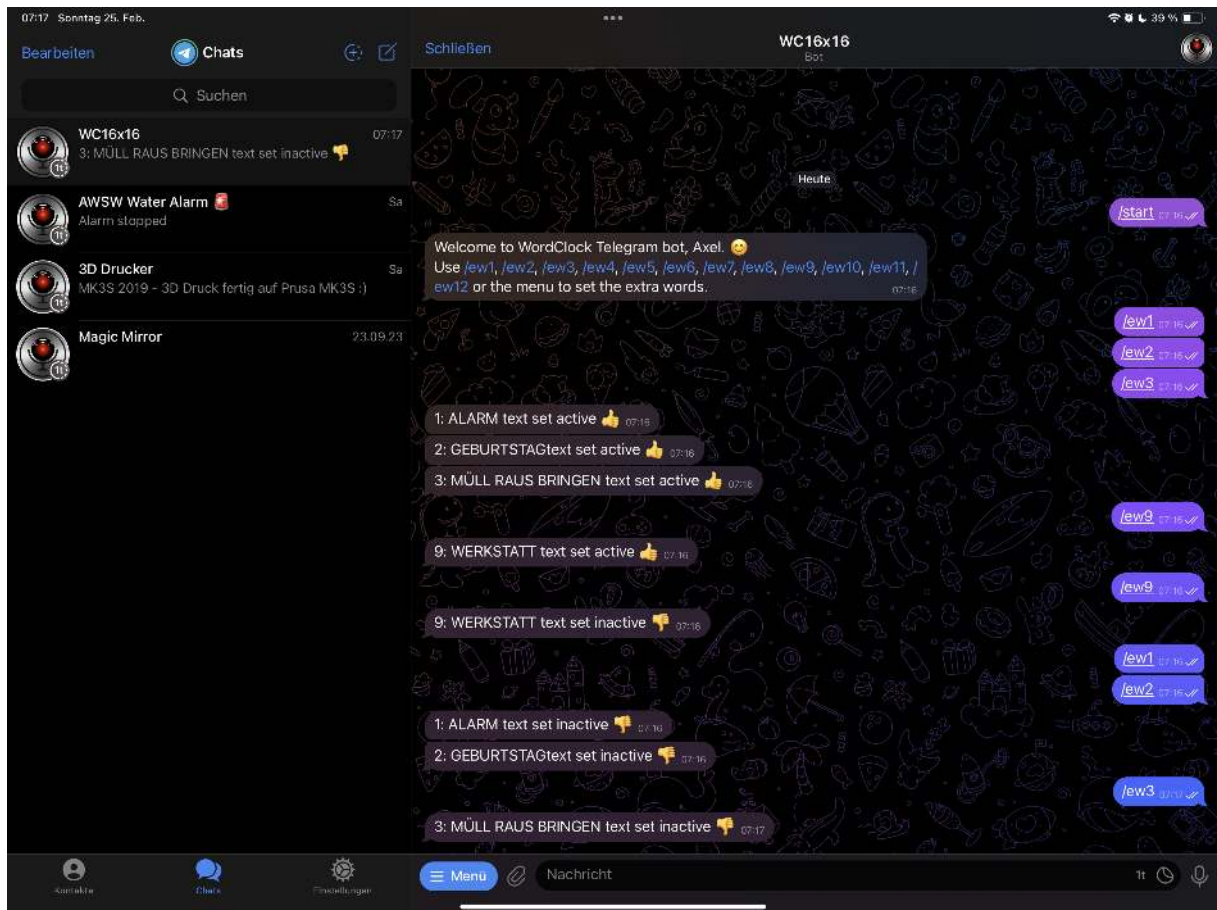
Extra words usage:

1.: Telegram:

- Create the Telegram bot as described step by step [here](#). Simply perform all the named steps from that list and you get the needed values and get the bot created and running.
- In addition you can setup the already preconfigured menu list in Telegram to help you with the usage of the bot.
- Add the in the generated Telegram bot token key and your chat id during the initial setup of the WordClock. So if you already attached WordClock to your WiFi without adding the Telegram setting, you will need to use the “Reset WiFi” button in the web configuration portal

and then setup WiFi, your language and the Telegram settings during the initial WordClock setup again.

- Type `"/start"` to get the list of available commands in Telegram.
- With the commands `"/ew1"`, `"/ew2"` to `"/ew12"` (DE) or `"/ew9"` (all other languages) or use the menu for the bot mentioned above during the Telegram setup to start the same commands via the menu in the Telegram app.



2.: WordClock internal web server commands to integrate into your smart home:

- You can control the following WordClock functions from your smart home setup:
 - Change time color
 - Change background color
 - Change intensity value (0 to the maximum limit set in the settings)
 - Set extra words color
 - Set extra words on or off
 - Get the status of an extra word
 - Get the status of all above named values (colors and on/off value) at once
- Examples and details can be found by opening this url from the internal web config in the section "Smart home control via web URLs" —> "Usage hints and examples" or by opening this url directly: "http://Your-WordClock-IP-address:2023".

Telegram support:

Use Telegram support (change forces restart)



React to your own Telegram ID only



Send notifications about messages from unknown users



Show your own Telegram bot token and chat id



Smart home control via web URLs:

About note

Control WordClock from your smart home environment via web URLs.

Functions

You can configure colors, the LED intensity and the extra words and get the status of each named element.

Usage hints and examples

<http://192.168.178.9:2023> or <http://esp32-A2B6EC:2023>

Update:

Activate update mode

Activate update mode

Update URLs

<http://192.168.178.9:8080> or <http://esp32-A2B6EC:8080>

Download newer software updates here

<https://github.com/AWSW-de/WordClock-16x16-LED-matrix>

WordClock web configurations and query options examples:

General:

<http://192.168.178.9:2023> --> Shows this text

Get the status of the WordClock at a glance:

<http://192.168.178.9:2023/status> --> Show the status of all extra words and the color for the background and time texts

Configure the WordClock colors for background and time texts:

<http://192.168.178.9:2023/config?R-Time=0&G-Time=255&R-Back=255&G-Back=0&B-Back=0&INTENSITY=25&INTENSITYviaWEB=1> --> Configures time texts to blue and the background to red. Intensity is set to 25 and sets this intensity value as master setting<http://192.168.178.9:2023/config?R-Time=0&G-Time=0&B-Time=255> --> Configures time texts to blue<http://192.168.178.9:2023/config?R-Back=255&G-Back=0&B-Back=0&INTENSITY=5&INTENSITYviaWEB=1> --> Configures the background to red.

Intensity is set to 5 and sets this intensity value as master setting

<http://192.168.178.9:2023/config?INTENSITY=0&INTENSITYviaWEB=1> --> LED intensity is set to 0 and sets this intensity value as master setting, which will turn the display off...<http://192.168.178.9:2023/config?INTENSITY=25&INTENSITYviaWEB=0> --> LED intensity is set to 25 and sets this intensity value as NOT master setting anymore, which will switch back to the in the web configuration set intensity parameters...

NOTE: The option INTENSITY has to be used together with the option INTENSITYviaWEB. This will ensure, that the automatic Day/Night-mode and the intensity setting that can be set in the internal WordClock web portal is not used at the same time and get disabled.

<http://192.168.178.9:2023/intensity> --> Show the status of the value INTENSITYviaWEB parameter

Set extra words:

<http://192.168.178.9:2023/ew/?ew1=1> --> Turns extra word 1 to on<http://192.168.178.9:2023/ew/?ew1=0> --> Turns extra word 1 to off<http://192.168.178.9:2023/ew/?ew9=1> --> Turns extra word 9 to on<http://192.168.178.9:2023/ew/?ew9=0> --> Turns extra word 9 to off<http://192.168.178.9:2023/ew/?ew3=1&R=0&G=0&B=255> --> Turns extra word 3 on and sets it to blue color<http://192.168.178.9:2023/ew/?ew3=0> --> Turns extra word 3 to off<http://192.168.178.9:2023/ew/?ew1=1&ew2=1&ew3=1&ew4=1&ew5=1&ew6=1&ew7=1&ew8=1&ew9=1&ew10=1&ew11=1&ew12=1> --> Turns ALL extra words to on<http://192.168.178.9:2023/ew/?ew1=0&ew2=0&ew3=0&ew4=0&ew5=0&ew6=0&ew7=0&ew8=0&ew9=0&ew10=0&ew11=0&ew12=0> --> Turns ALL extra words to off

Get status of a single extra word:

<http://192.168.178.9:2023/ewstatus/?1> --> Status of extra word 1<http://192.168.178.9:2023/ewstatus/?3> --> Status of extra word 3<http://192.168.178.9:2023/ewstatus/?9> --> Status of extra word 9

Set all extra words off until midnight:

<http://192.168.178.9:2023/resetew1> --> Set all extra words off until midnight<http://192.168.178.9:2023/resetew0> --> Set all extra words outputs active again<http://192.168.178.9:2023/resetewstatus> --> Get the status of this switch

Smart Home environment integration:

- If you use similar smart home control environments like **HomeBridge** and **NodeRed** like I do, you can add such simple configurations to plugins like “homebridge-http-switch” in HomeBridge to control the extra words from your environment systems with the use of calendar plugins like “homebridge-calendar” automatically. If you use HomeAssistant or similar programs you will get the idea how to apply this to your used systems...
- **Side note:** If you ask yourself why I use these 2 systems instead of whatever you are using, well the WordClock project was created by me after I already used a smart **MagicMirror** for years before. The with HomeBridge and NodeRed created functions and dashboards were used in this really still fascinating smart mirror project already...
- Code and settings within the HomeBridge plugins “homebridge-calendar” and “homebridge-http-switch” for extra word “ew6” which is “Formel1” in German:



Plugins

Suche nach



Einstellungen: homebridge-calendar



Bitte lese die Plugin Dokumentation, um das Plugin korrekt zu konfigurieren.

Calendar

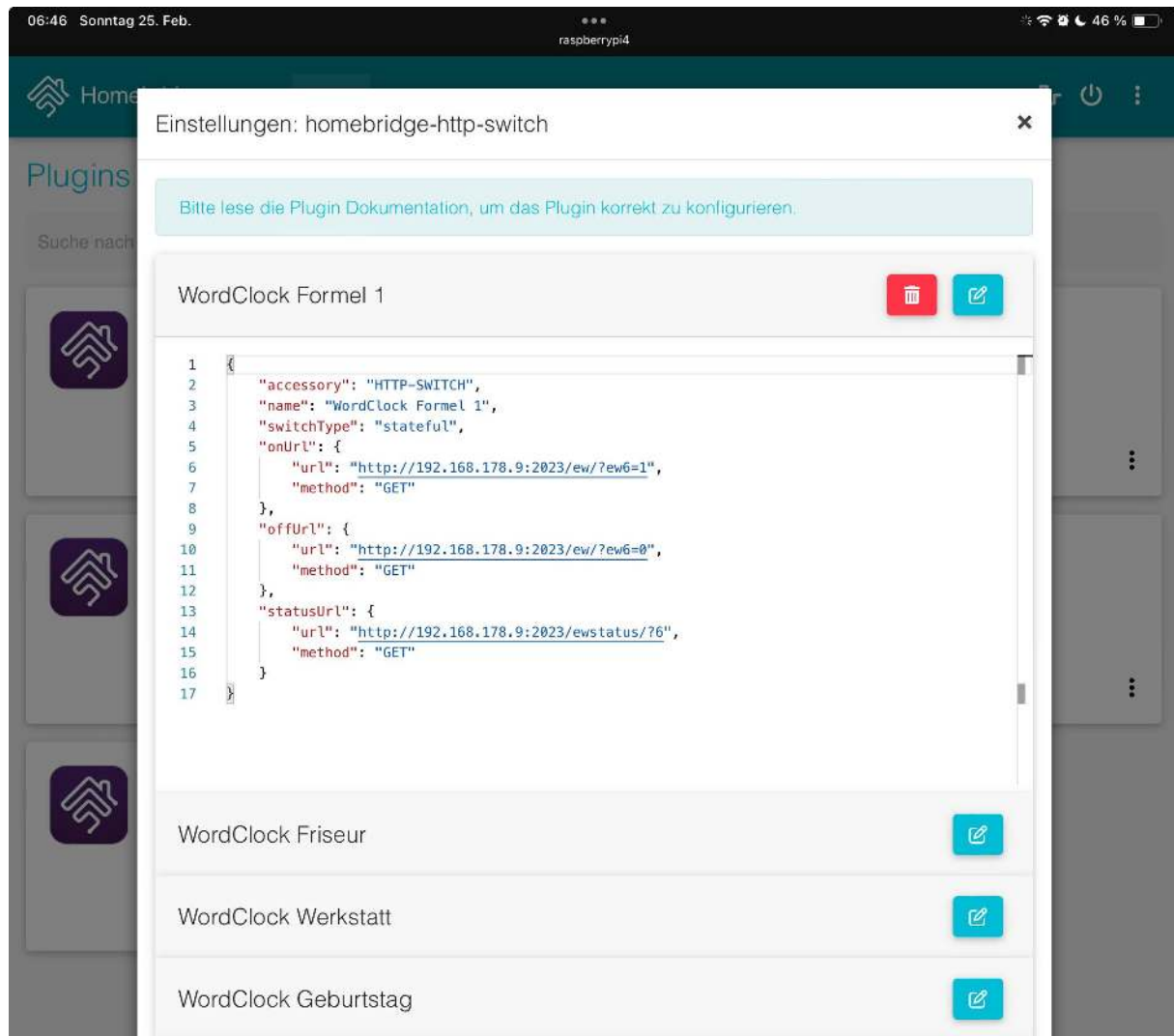


```
1 {
2   "platform": "Calendar",
3   "calendars": [
4     {
5       "name": "Regeltermine",
6       "url": "webcal://p101-caldav.icloud.com/published/2/M",
7       "pollingInterval": 5,
8       "offset": "-16h",
9       "sensors": [
10        "Gelber Sack", "Friseur", "URLAUB", "BMW Service Termin"
11      ]
12    },
13    {
14      "name": "Serien",
15      "url": "webcal://p101-caldav.icloud.com/published/2/M",
16      "pollingInterval": 5,
17      "offset": "-16h",
18      "sensors": [
19        "F1:"
20      ]
21    }
22  ],
23 }
```

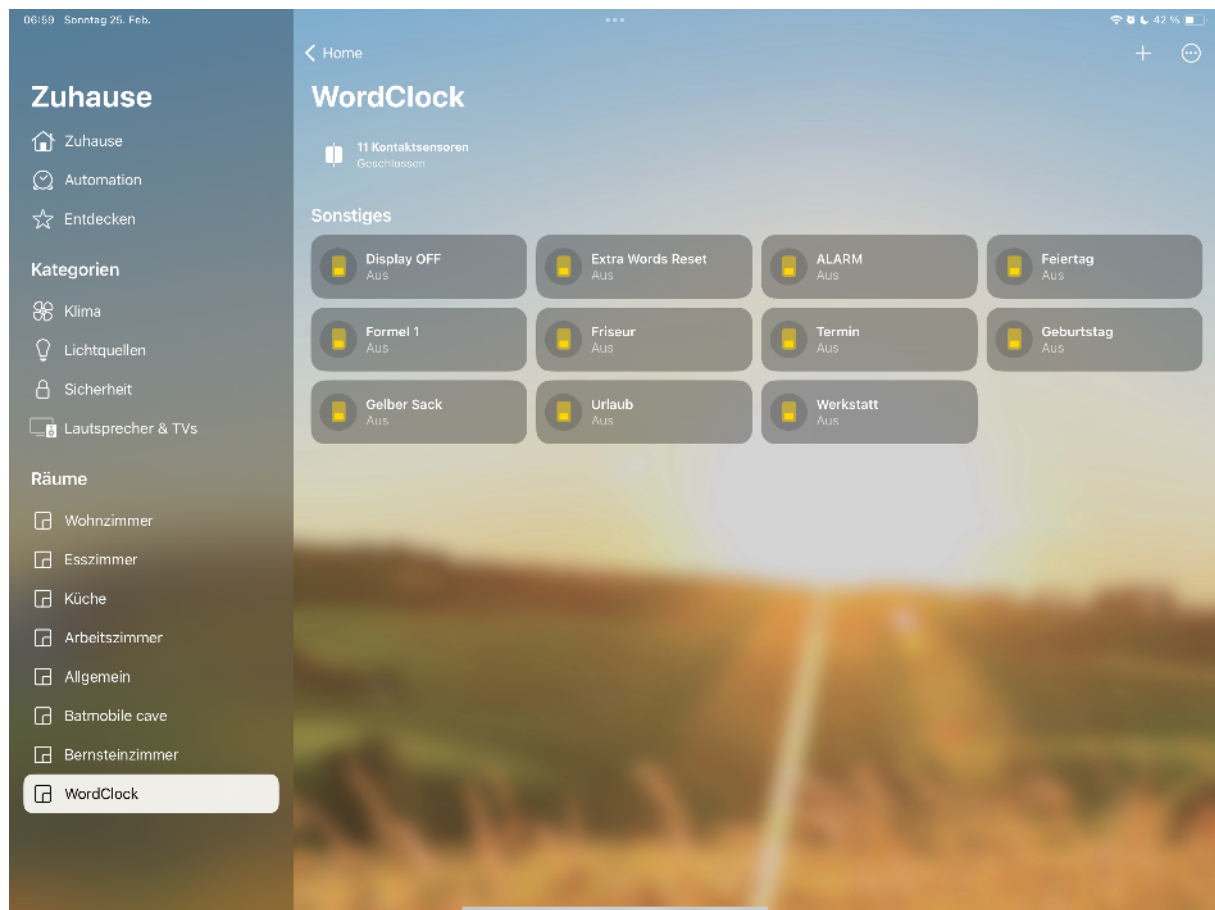
SCHLIESSEN



SPEICHERN



- **Side note:** In addition you will find these created switches in Apple HomeKit too and of course can control them in your HomeKit app as well as via Siri and within other automation workflows too:



- Settings in the NodeRed plugin “HomeBridge” named exactly “node-red-contrib-homebridge-automation” and the in this plugin included controls “hb-event” and “hb-control”:

06:51 Sonntag 25. Feb.

raspberrypi4

Node-RED

Nodes filtern

Flow 1

inject

debug

complete

catch

status

link in

link call

F1:

Friseur

Geburtst

Gelber S

URLAUB

Einstellungen

Schließen

Ansicht

Installierte Nodes

Installation

Homebrid

1 / 14

node-red-contrib-homebridge-automation

0.1.11

5 Nodes

In Gebrauch

hb-conf

hb-event

hb-resumé

hb-status

hb-control

In Gebrauch

06:35 Sonntag 25. Feb.

raspberrypi4

Node-RED

Nodes filtern

Flow 1

inject

debug

complete

catch

status

link in

link call

link out

comment

F1:

Friseur

Geburtstag

Gelber Sack

URLAUB

Feiertage

function

function

function

function

function

function

WordClock Formel 1

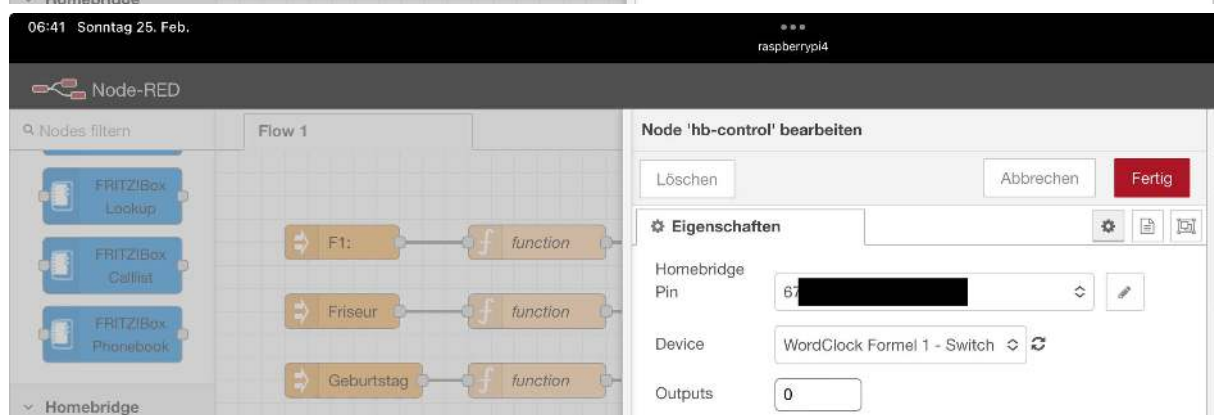
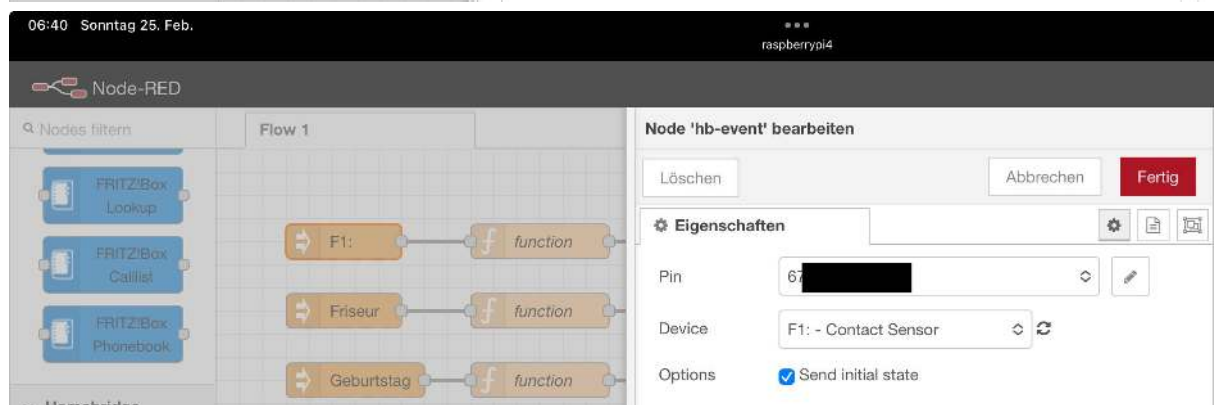
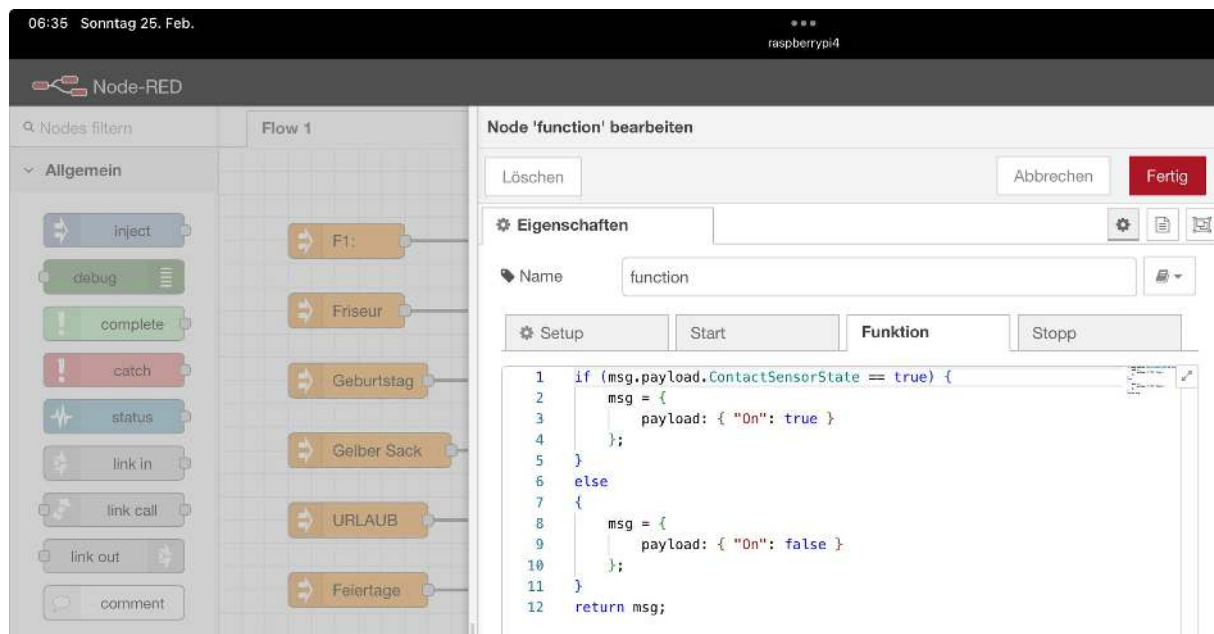
WordClock Friseur

WordClock Geburtstag

WordClock Gelber Sack

WordClock Urlaub

WordClock Feiertag



Alternative Smart Home environment integration via Apple shortcuts:

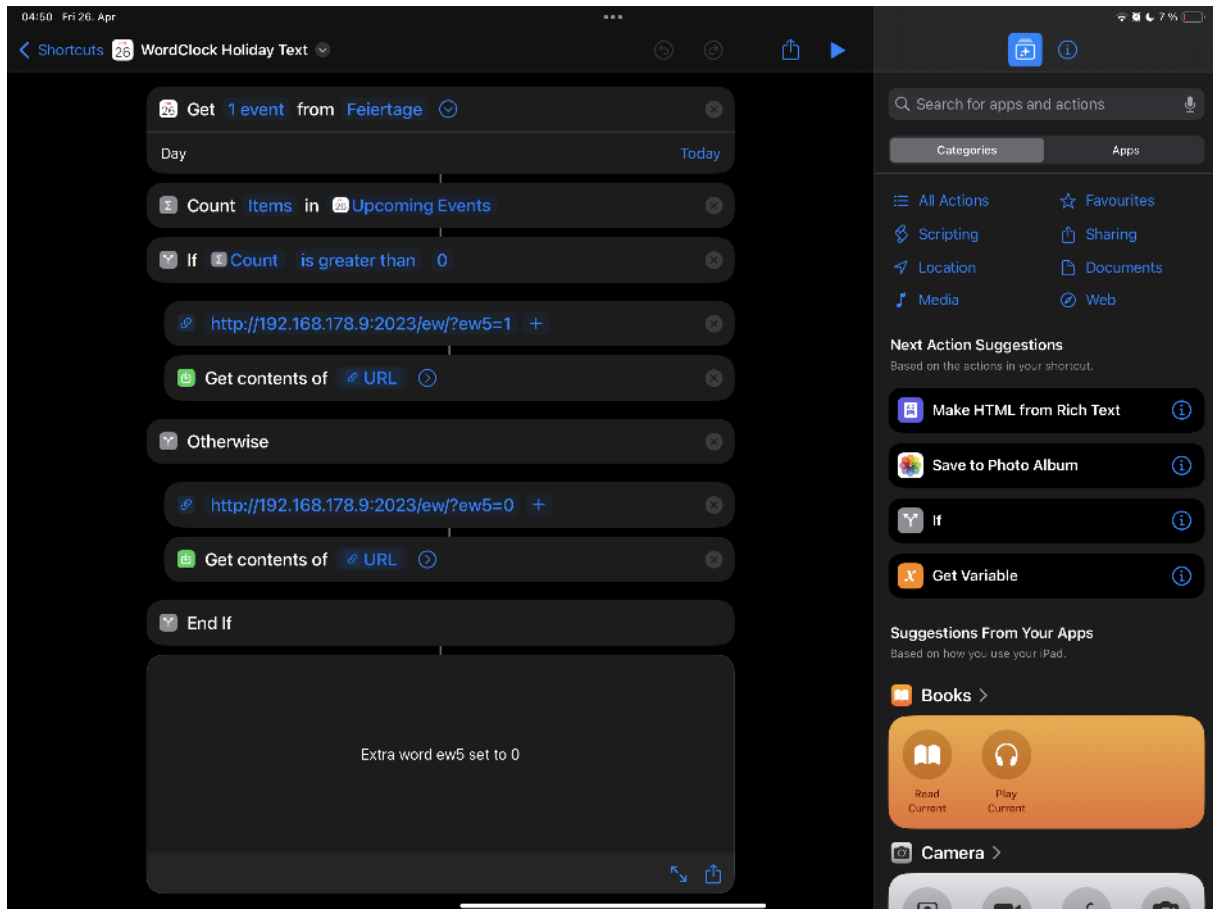
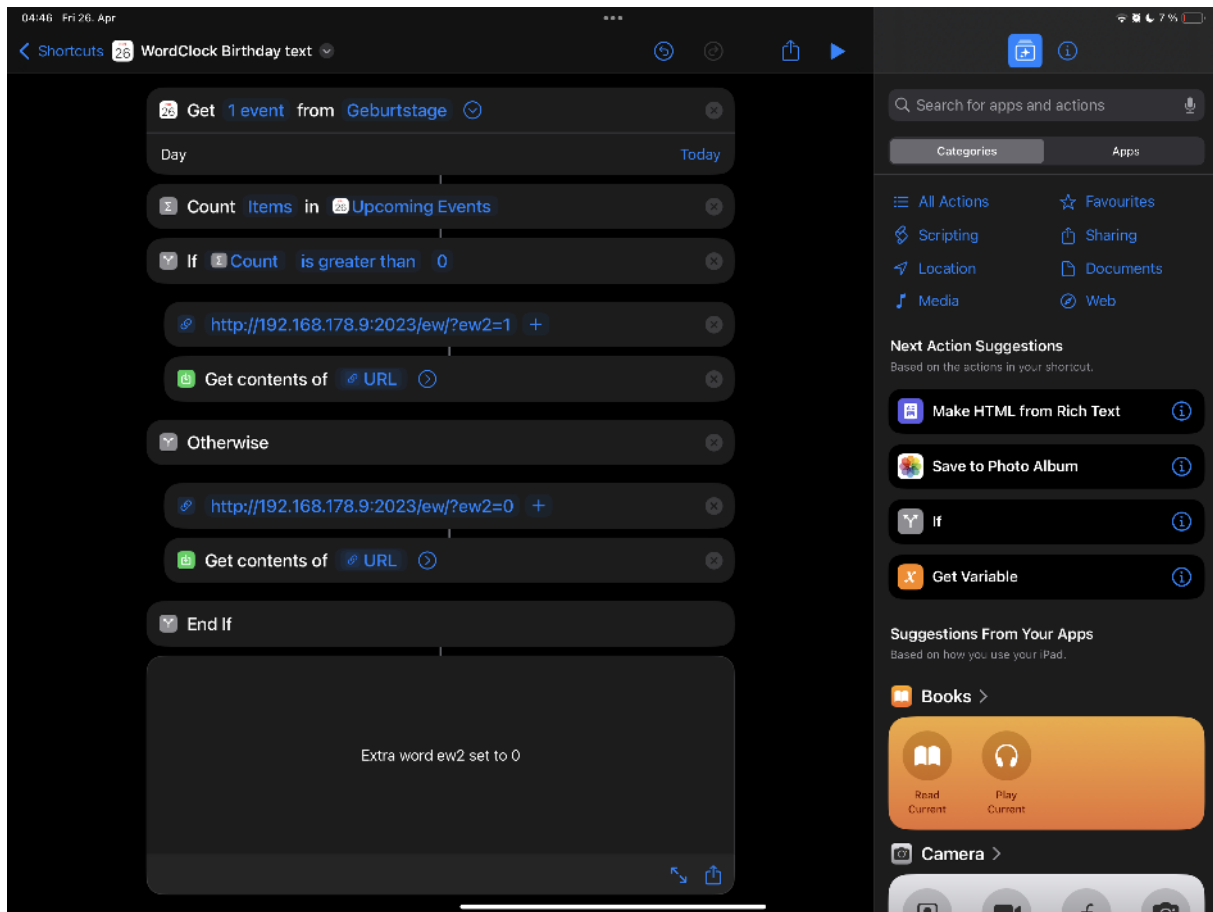
In case you don't use such smart home environment devices like mentioned above, you can simply add the named extra word URLs to the "Shortcuts" of your Apple iPhone or iPad and run them on a fixed time of each day. This might work similar on other devices like Android, but I use Apple devices only. So here you will find a simple example for 2 extra words in German "Geburtstag" which is "Birthday" and "Feiertag" which is "Holiday" in English language.

Steps to follow in the shortcuts app usage:

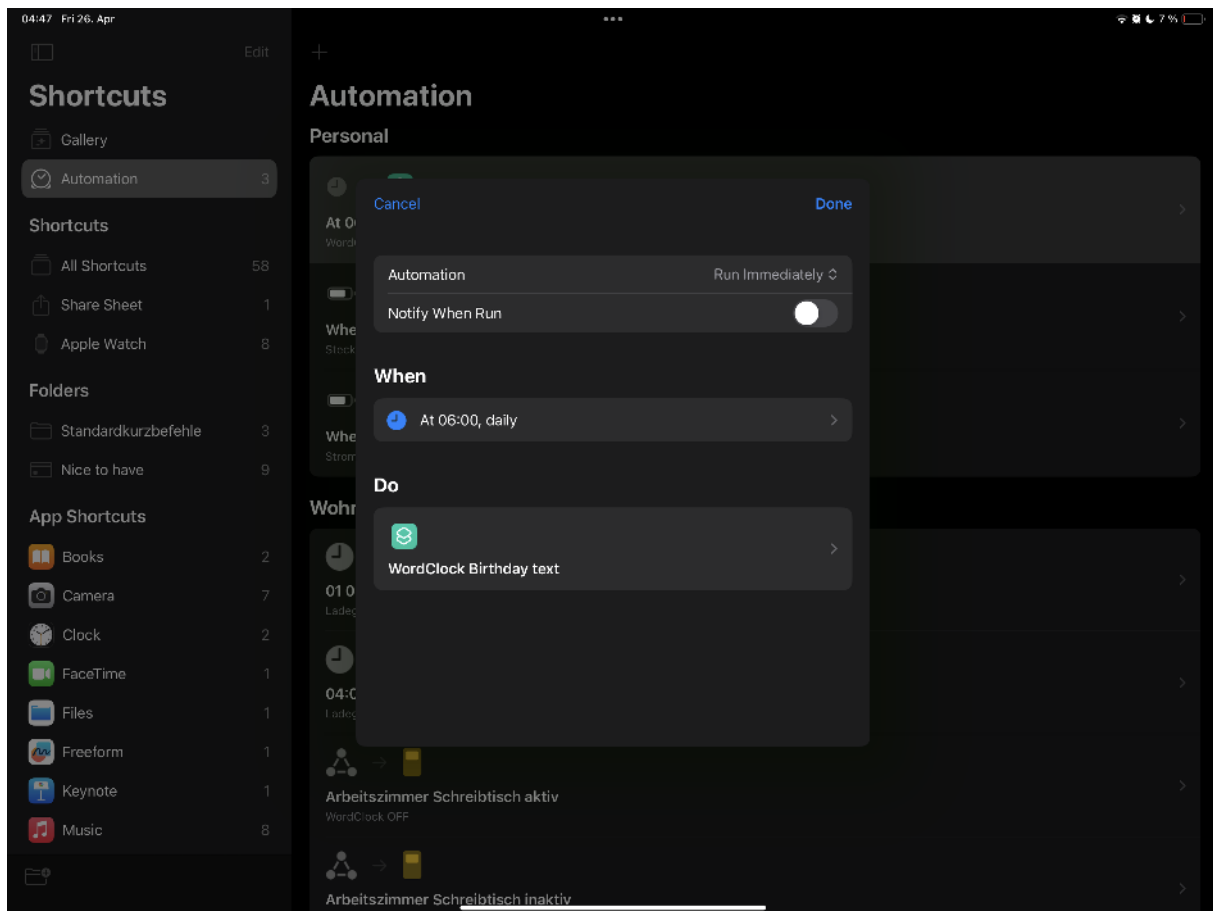
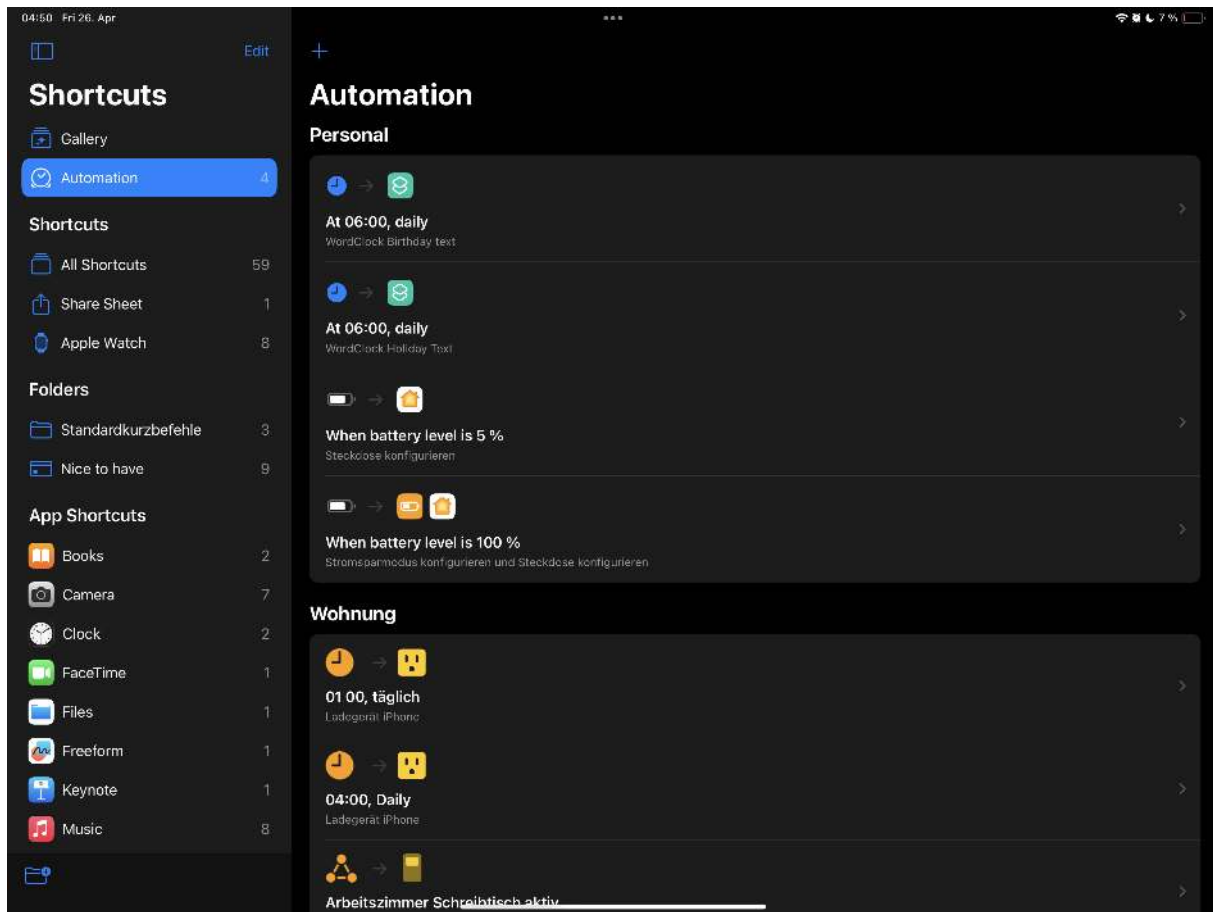
- Get the numbers of your extra words you want to setup. You can find these numbers in the WordClock configuration portal in front of the extra words:
- Geburtstag = Birthday = 2
- Feiertag = Holiday = 5



- Use the shortcuts app to add a new shortcut like shown below for the 2 extra words and combine them to your separate calendars. One calendar for your birthday events and one calendar for the holidays to keep this simple.
- The shown IP-address will be needed to adjust to your local environment value for your WordClock. You can use the hostname instead of the IP-address of your WordClock too if you like of course.



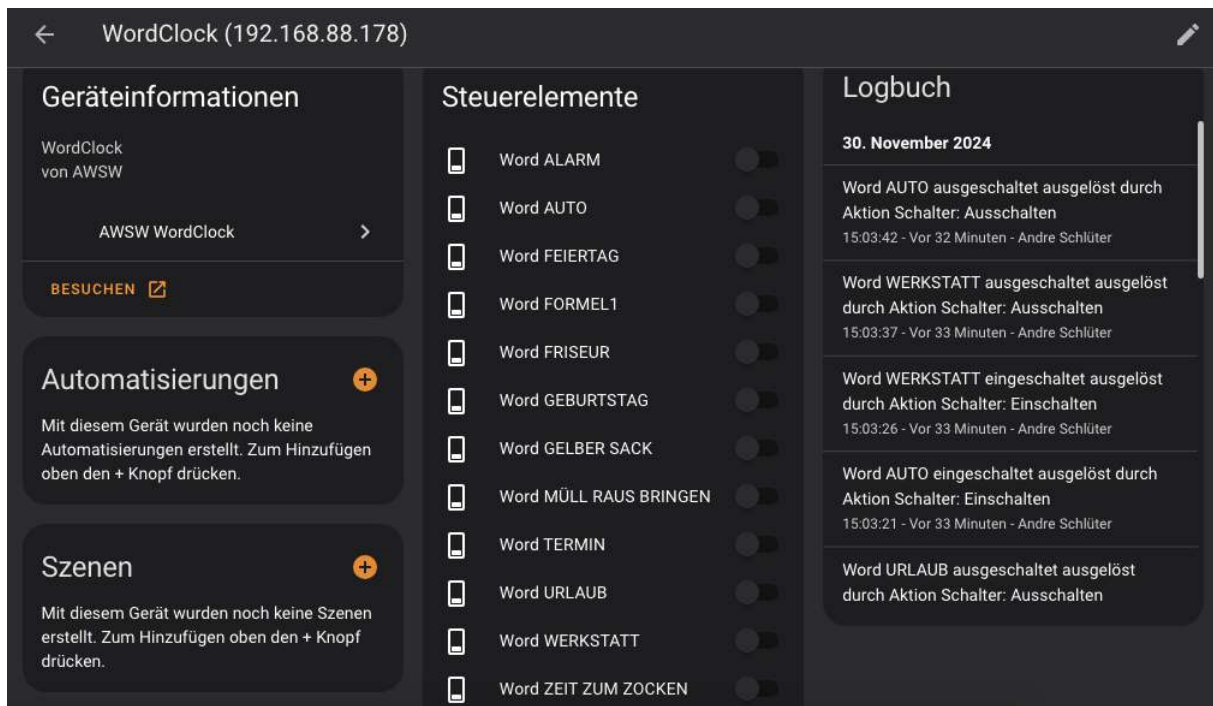
- Add a local automation in the shortcuts app like shown below to a fixed time and set the execution to be without additional apply to run unattended in the background.



With that your iPhone or iPad will set the texts on your WordClock based on the calendars. =)

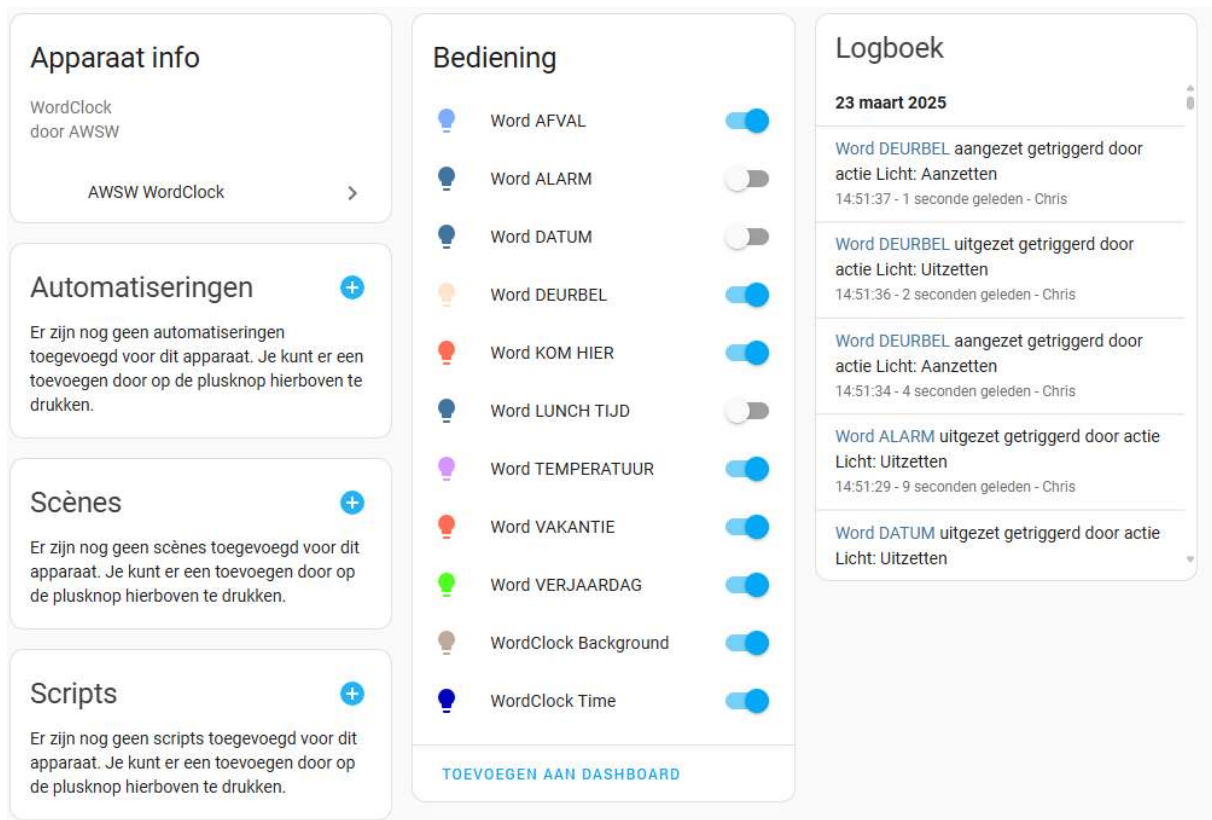
WordClock Integration for Home Assistant:

- The user bluenazgul added a solution for Home Assistant [here](#).



Note: Photo by blueszgul

- KRIZ-R made an even more advanced version of this using the RGB query of the extra words introduced with WordClock software version V4.7.1:
- Link: https://github.com/KRiZ-R/HA_AWSW_Wordclock



Note: Photo by KIRZ-R

FAQ:

- Q: “ **Can I use a smaller power supply like 5V/1A or 5V/2A?**”
 - A: Do NOT use 5V/2A or even 5V/1A ones to avoid damage to your environment!
- Q: “ **Why is it not possible to use a USB-C power supply directly?**”
 - A: The honest answer is, that it was planned to be used with a USB-C power supply directly, but it simply does not work. Most USB-C power supplies “communicate” with the plugged in device and then turn their power on. This is not possible with the WordClock. So this is why a USB-A power supply with a USB-A-to-USB-C cable is required. You will find 2 very good ones in the parts list.
 - A: There is an advantage to this anyway. Because such USB-A-to-USB-C cables seem to have a better standard, they work much better than the previously used Micro-USB cables.
- Q: “ **Can I use my old ESP8266 boards instead of the named ESP32?**”
 - A: No ESP8266 usage possible, because of its low specs. It is not compatible too.

- Q: “ **Why isn’t the circuit done like with the previous models over the USB cable only?**”
 - A: That often lead to some poor grilled ESP32 boards when the LED intensity was set to high and too much current was routed through the ESP32 to the LED matrix. That also seems to happen due to user reports often too, when other code sketches like WLED or others were “tested”... After some time or directly depending on the set color and intensity this caused some of these boards to die. As this seems to be an issue for some users and even I never experienced this myself, I wanted to make the circuit better.
 - A: The 2nd reason is that I wanted to try a solder free variant and so it was even more simple to save these poor little boards from dying on their job. =)
- Q: “ **Why was the switch from the ”D1 mini ESP32” board done to the new ESP32?”**
 - A: The 1st reason is that during the runtime of the previous WordClock models I heard of several problems with these D1 mini boards from users, which mostly I did not face. What I could find on very few of these boards was the “Brownout detector” error message which prevented these boards from booting. Some of them worked for days and even weeks. Then I needed to replace the Micro-USB cable to a shorter one and for any reason then it worked most of the times. A few days later the same board worked with the “old” cable again without any problems.
 - A: The 2nd reason is that the new chosen NodeMCU ESP32 board has a USB-C connector and I had often issues with the old Micro-USB cables and ports on the old ESP32.
 - A: The 3rd reason is that this board can be easily used to reach the solder free variant and is possible to screw to the custom made mounting socket in the lid.
 - A: The 4th reason is that I got slightly better WiFi connection and boot operations with these new boards compared to the old ESP32.
 - A: The 5th and most important reason for this new 2024 model was, that the D1 mini ESP32 cannot be powered without the Micro-USB cable usage. It seams to have a design issue that it cannot be powered from the VIN/5V and a GND pin on its board. I searched many forums and pages to get it working over such a pin connection, but it was not possible. So the new NodeMCU ESP32 will be used here.
- Q: “ **Where do I need to set the color changes and how are these changes done?**”
 - A: See printing instructions below please. You will find a manual and details.

- A: Consider using Prusa Slicer even for your non-Prusa 3D printer too. =)
- Q: “ **Do I really need to print with black filament?** ”
 - A: Yes, to avoid light shining through to the other letters this is required or lets say at leased it is required as light blocker. You can add other, lighter colors to the first front part layers if you wish to, but you have to have this light blocking black layers. For the matrix part of the front it is mandatory to use black too.
- Q: “ **Can I use transparent PLA instead of the named white filament for the diffusor?** ”
 - A: Tests showed that this is not ideal, because such transparent filament will route light to other cells of the matrix that should not be lit at that time or in that color.
- Q: “ **I cannot read the on the display shown IP-address. What is my IP-address?** ”
 - A: Just step away 1-2 meters from WordClock during startup to be able to read it.
- Q: “ **How can I setup the Telegram option?** ”
 - A: If you want to use the Telegram function, follow the instructions in the settings.h file. Add the BOTtoken and your CHAT_ID value in the settings.h file and save it.
 - Optionally add the menu for the Telegram bot like mentioned in the settings.h file.
- Q: “ **Why is the software not provided as self to compile code to run in Arduino IDE?** ”
 - A: Although I created scripts for setting up the Arduino IDE, automatically download and install needed libraries and spend a lot of time to give you a setup option as easy as possible, it seems to be still way to complicated for many users. Thats why you have the new script to download and flash the code for you now as precompiled .BIN files now. I had so many messages from users each week that did not (fully) follow the already quiete simple instructions and the only possible way for me is to make this as easy as possible and not provide the more complicated way anymore.
 - A: By giving you the precompiled code you can be sure that this is tested and will work with the named components, which should be a big benefit. =)
 - A: And YES, I know, that other ESP32 boards might not work with these precompiled code files, but I honestly have to say, that this is a ≤5€ or ≤5\$ problem and I perform this here as a hobby and cannot support many different ESP32 variants anyway... I guess you will understand this, that I cannot and will not spent more time on this when it can be fixed with buying such a cheap ESP32 fitting to the project.

- Q: “ In Offline Mode the from the device created WiFi access point requires a password, which I don’t know. What is this password and does the configuration page open automatically when connecting to this internal access point?”
 - A: “The password is ‘ **Matrix-16x16**’ and in case the configuration page doesn’t open up properly on your device (mostly heard of on Android devices) just connect to the WiFi and then open your browser on ‘**http://192.168.4.1**’. Once saved to your device the password should be stored and you not need to enter it again anymore. This password was added to make the device more secure for example if you want to use it on your work environment without access to a regular WiFi.”

Printing instructions:

- No supports needed for all parts.
- Using a brim was not needed in my tests, because I cleaned my textured sheet with dishwashing liquid carefully with a new sponge and then with isopropanol alcohol again on the printer to make the parts stick really good. If you face problems with adhesion feel free to add a 5mm brim to the front and lid part.
- 0.3 draft profile used. You can print in 0.2, but it was not needed for my printers.
- Textured sheet used:
 - If you use a textured sheet too, let it cool down completely after the print and wait for the part gets loose by itself. Do not pull on it to avoid ripping off the letters!
 - In my case it takes 45-60 minutes after the print has ended and the part just lays loose on the print bed.
 - Do not touch the print bed and clean it with Isopropanol alcohol before printing.
- The option “ **Avoid crossing perimeters**” was set active for the front parts in Prusa Slicer.
- The 4 screws should be printed with 30-50% infill just in case you need to screw them in a little harder. I could use 15% infill good as well.
- **IMPORTANT:**
 - Two (2) color changes in the front part from black to white and to black again are required at about 1.2mm and 2.4mm height to create the diffuser layer. Why “at about”? Well it depends on your layer height:
 - 0.2mm = Changes at 1.2mm + 2.4mm
 - 0.3mm = Changes at 1.1mm + 2.6mm
 - Color changes are done in this order:
 - 1st color = black = layer 0 to 1.2/1.1mm
 - 2nd color = white = layers 1.3/1.2mm to 2.4/2.6mm

- 3rd color = black = layers 2.5/2.7mm to the end of the print
- To read how such a color change is done, you will find a really good [article](#) here.
- If you use the .3MF files the color change is preconfigured for 0.3mm layer height.

Parts needed:

- 1x [16x16 LED matrix](#) (Use this one. The case is made to fit to it only!)
- 1x [NodeMCU ESP32 USB-C](#) (Use this one. The case is made to fit to it only!)
- 1x [USB power supply 5V / 3A](#) or e.g. the USB-A port from this [one](#) (No direct USB-C!)
- 1x [USB-A to USB-C cable \(5V, 3A\)](#) (No direct USB-C!)
- 1x [USB-C port \(5V, 3A\)](#)
- 3x [Wago 221-413 - 3 wire clamps](#)
- 3x [Jumper Wire 40x 20cm, Male-Female](#)
- 3x [M3x16 screws](#)
- 4x [M3x8 screws](#)

Tools needed:

- 1x [FDM 3D printer](#) and PLA filament in [black](#) and [white](#)
- 1x [Wire stripper](#)
- 1x [M3 allen key](#)
- 1x [side cutters](#) (to cut the cable ties and cables cleaner)
- 4x [cable ties](#)
- 1x [soldering iron set](#) (if you want to solder)
- 1x [3rd hand](#) (if you want to solder)



Wiring:

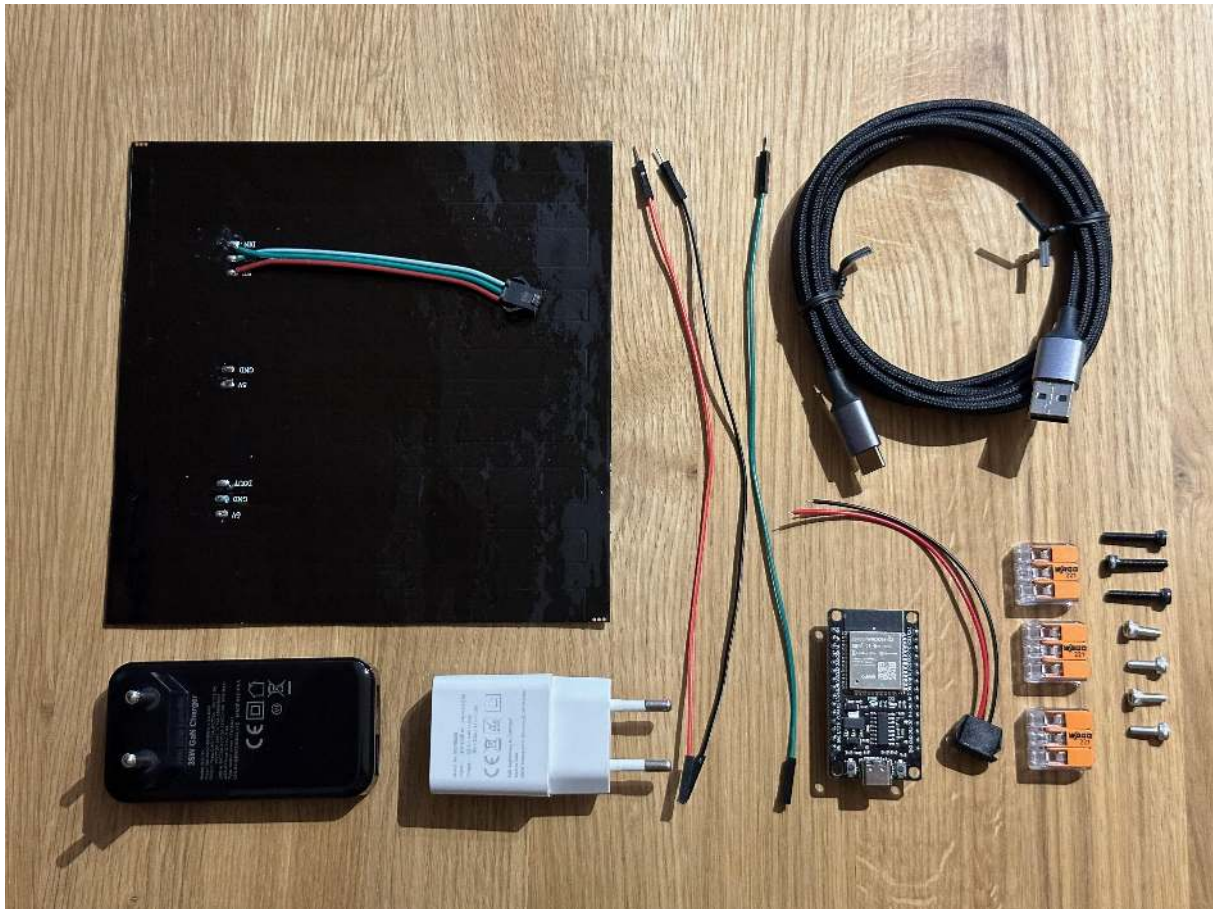
- Connecting the parts for this WordClock is as easy as possible.
- You may need to solder or connect solder free 3 connections only to the ESP32.
- The small holders in the lid can be used for a small cable ties to secure the cables.
- Have a look into the pictures how to arrange the LED matrix in the front part to ensure that the direction of the LEDs is correct.
- In case you want to solder, the only change compared to the solder free variant would be to solder the 3 wires to the ESP instead of using the named plugged cables. Details will not be described here due the connections are the same as for the solder free variant. The Wago clamps would be used in this case too, because they are really handy to use...
- Please check the table which pins are connected for the project as well as the pictures:

USB-C port:	ESP32:	LED matrix:	Wire color:
5V	VIN	5V	Red

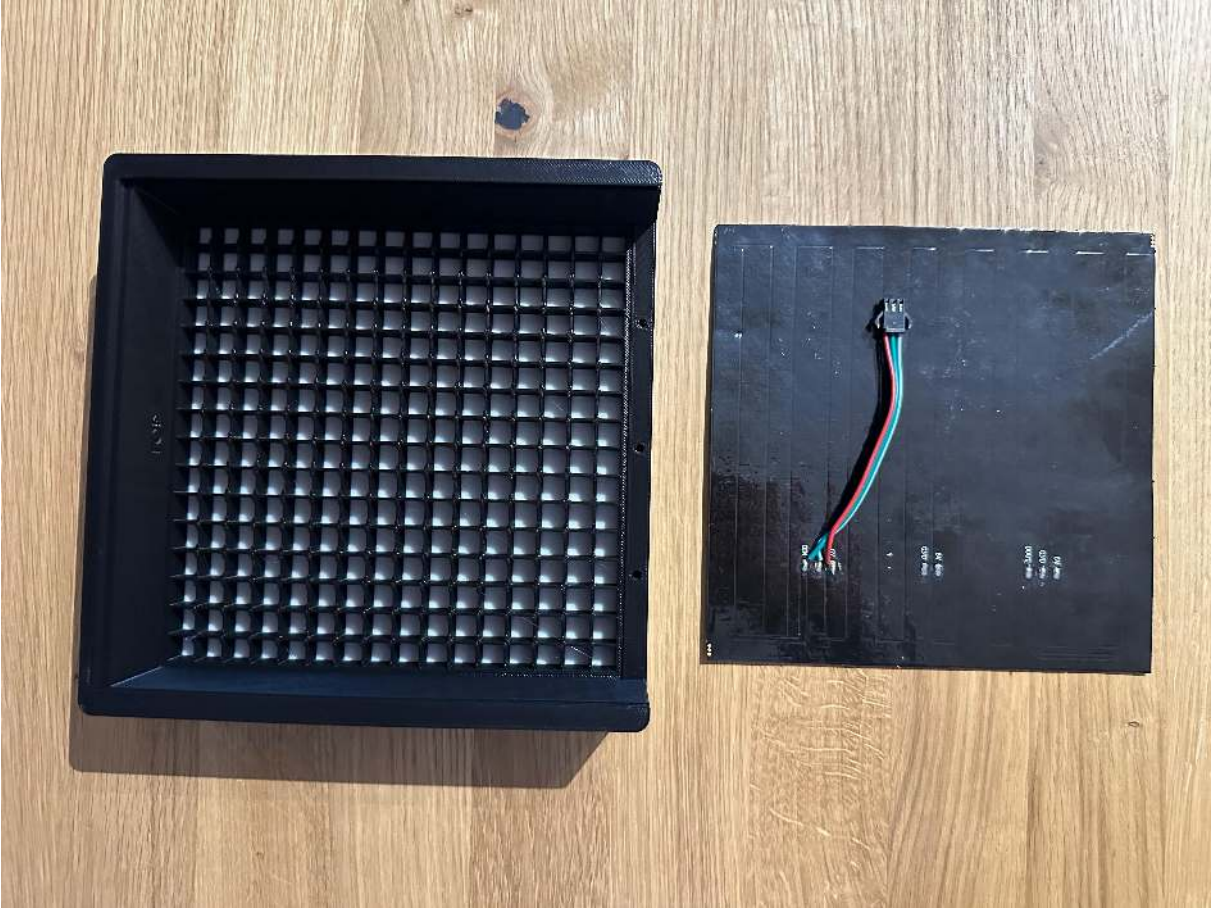
GND	GND	GND	Black or White
	D32	DIN	Green

Here is a step by step guide how to connect all the cables:

- With the help of the following pictures you will see how to wire the device, easy as 1,2,3:



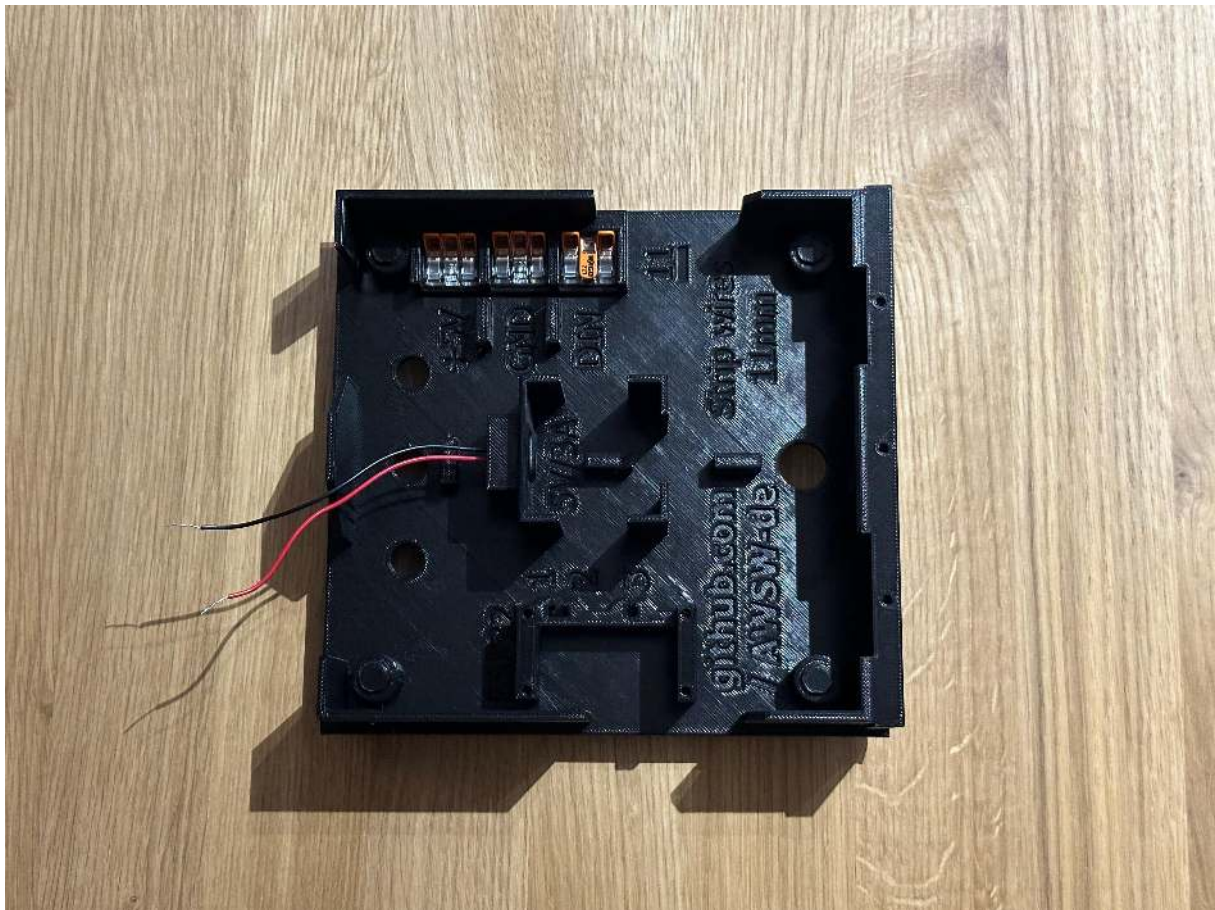
- Here you see all the required parts.



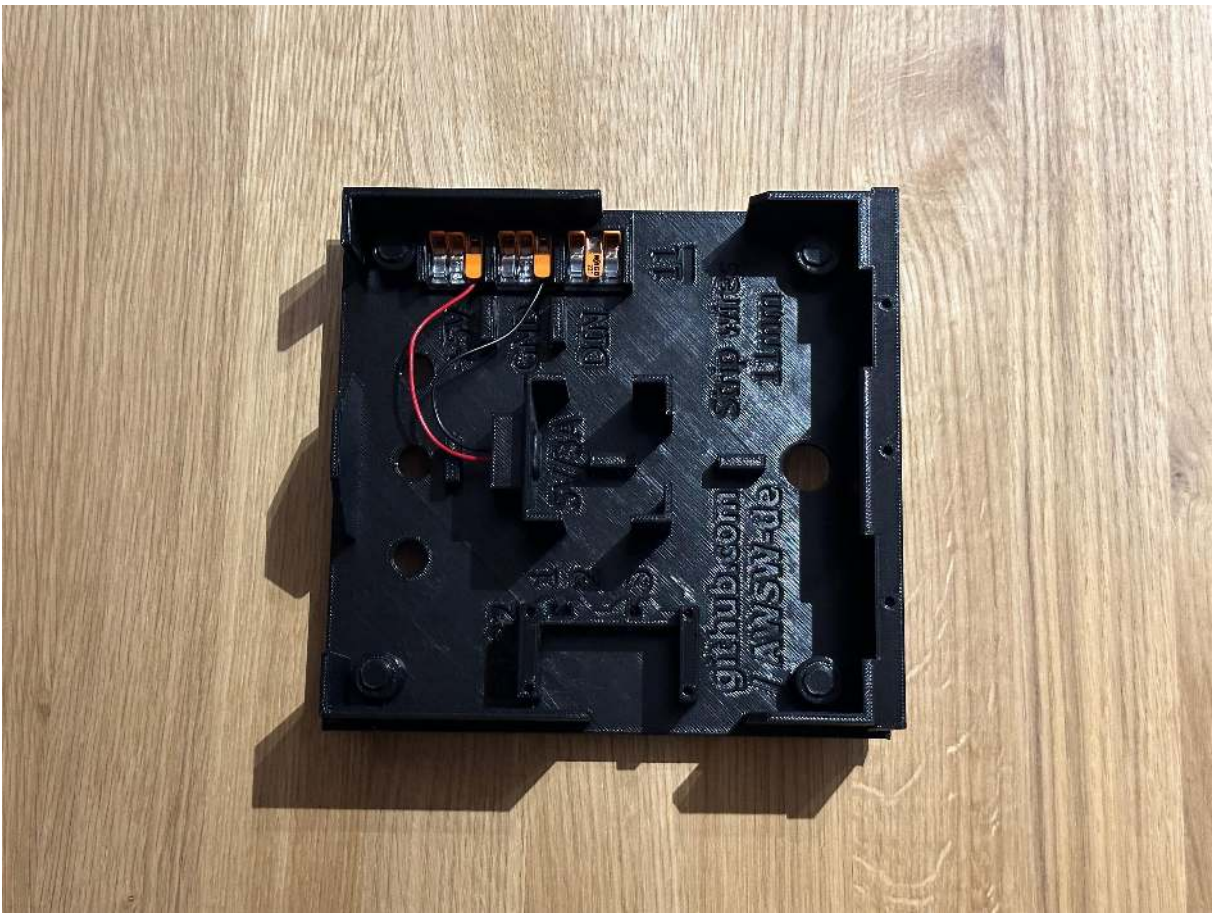
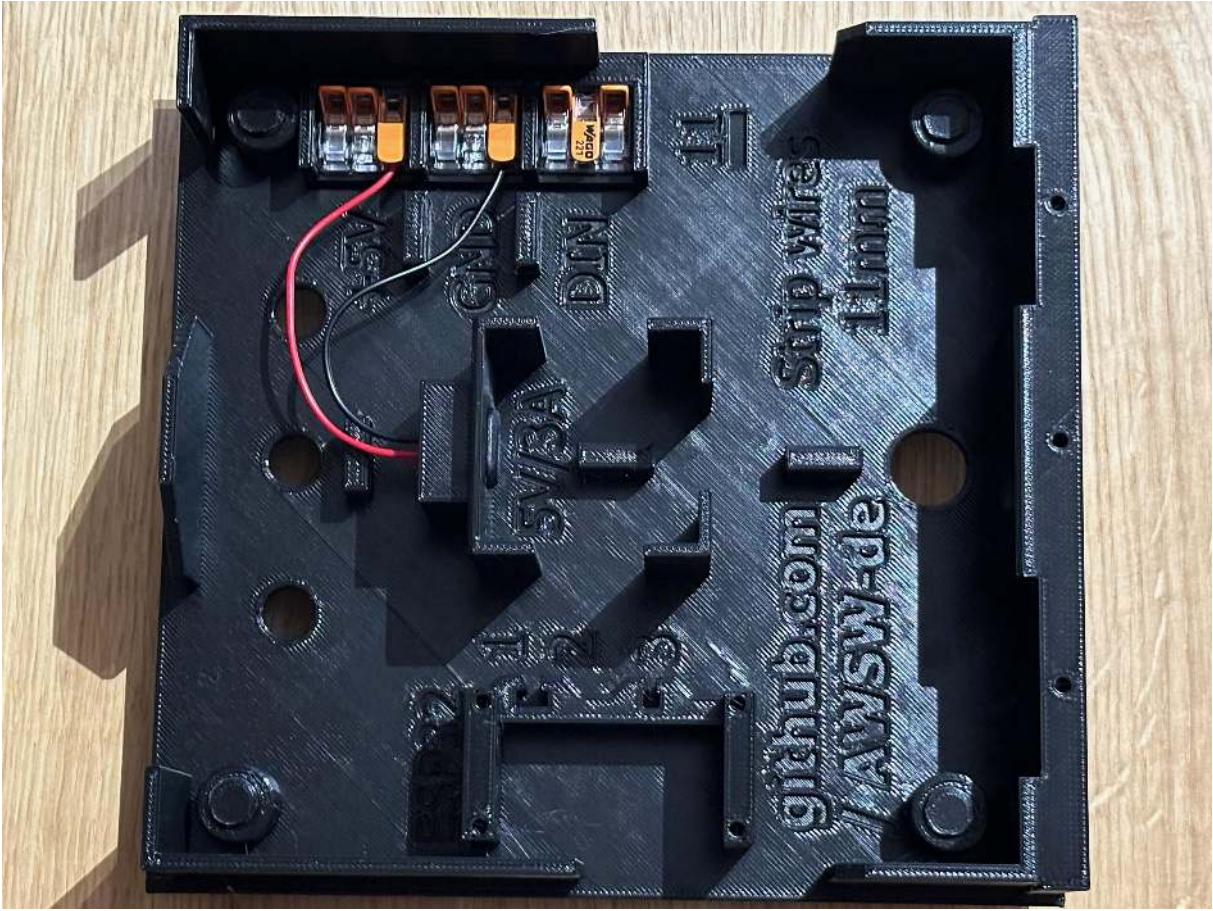
- Cut off the middle and lower cable pairs from the matrix to avoid electric short circuits. The cables at „DIN“ have to remain. The result is shown here.
- Add the LED matrix in the orientation as shown.



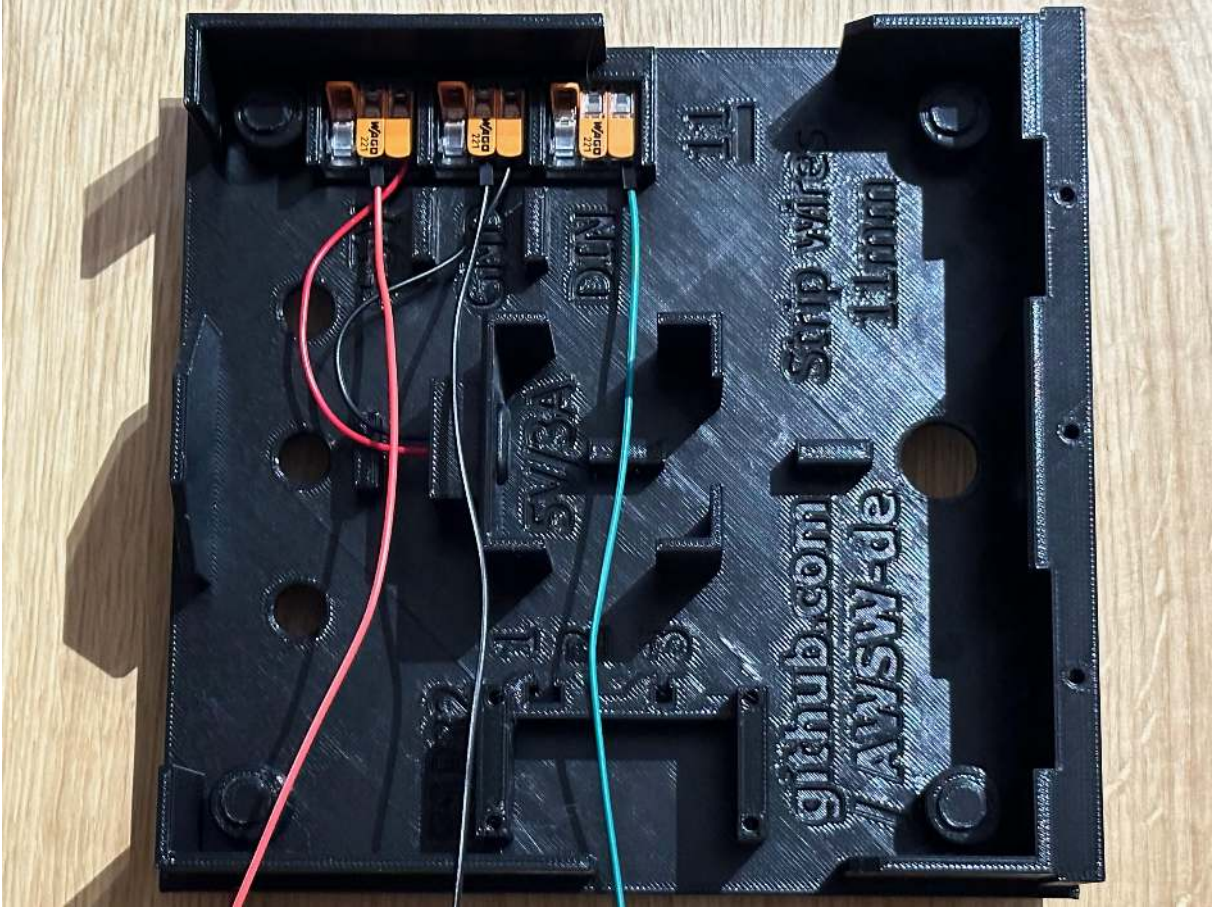
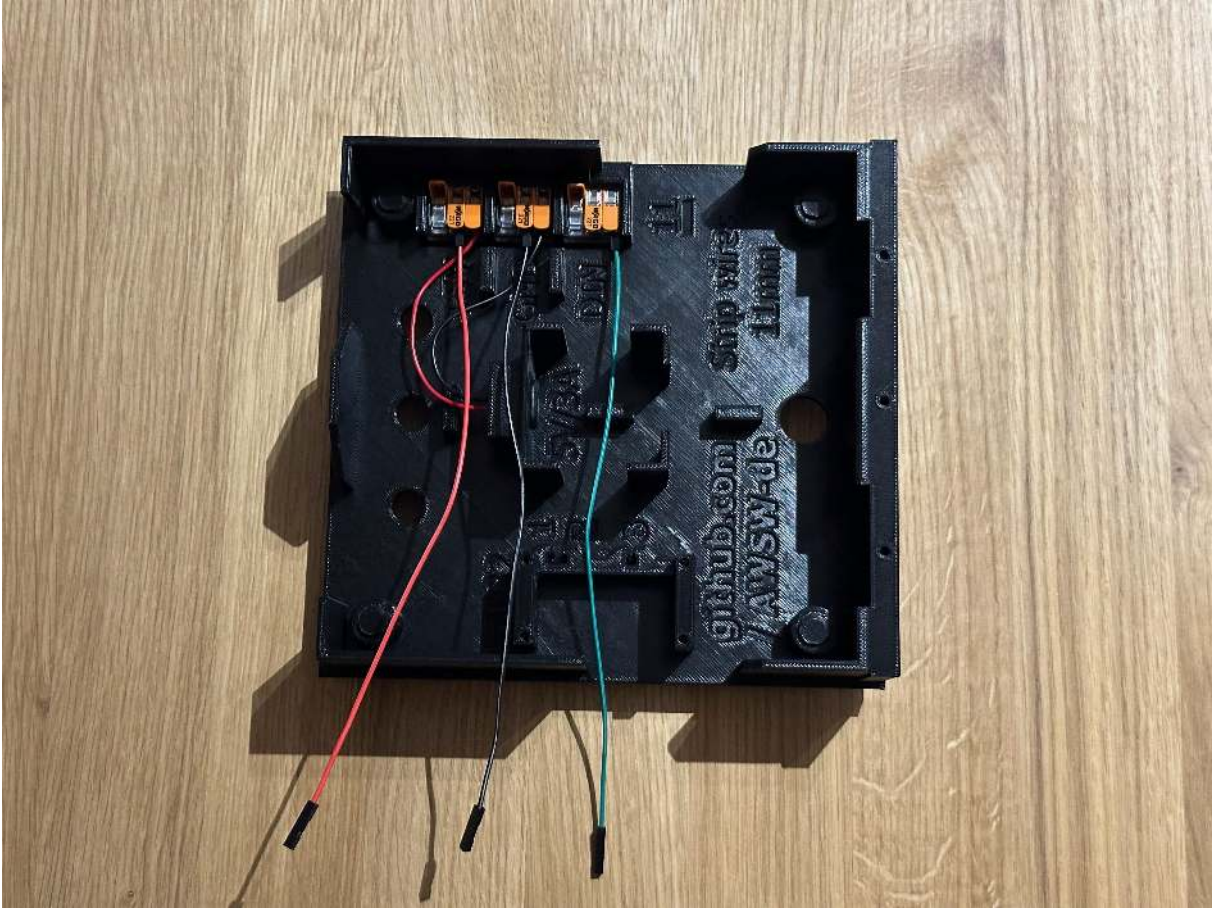
- Next strip the 2 USB-C port wires 11mm and push it into its place in the lid. It will click in place with a bit of the force. ;-)
- You will find a 11mm long mark in the case to help you finding the right strip length. See the upper edge in the picture above with the "11".



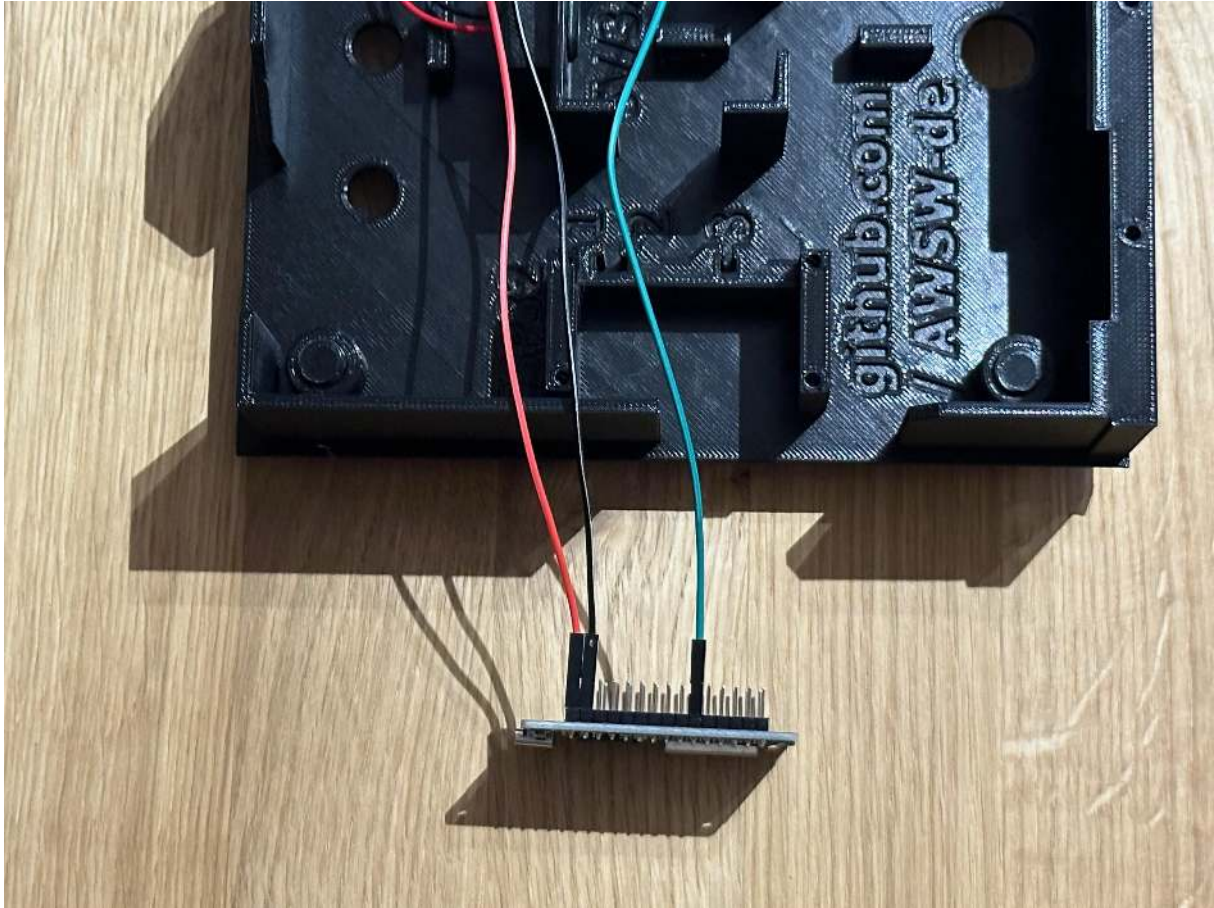
- Open the 3 Wago clamps and put them into their holders in the lid as shown.
- Remember the 1,2,3 under the clamps as:
 - 1 = red = 5V
 - 2 = black or white = GND
 - 3 = green = DIN



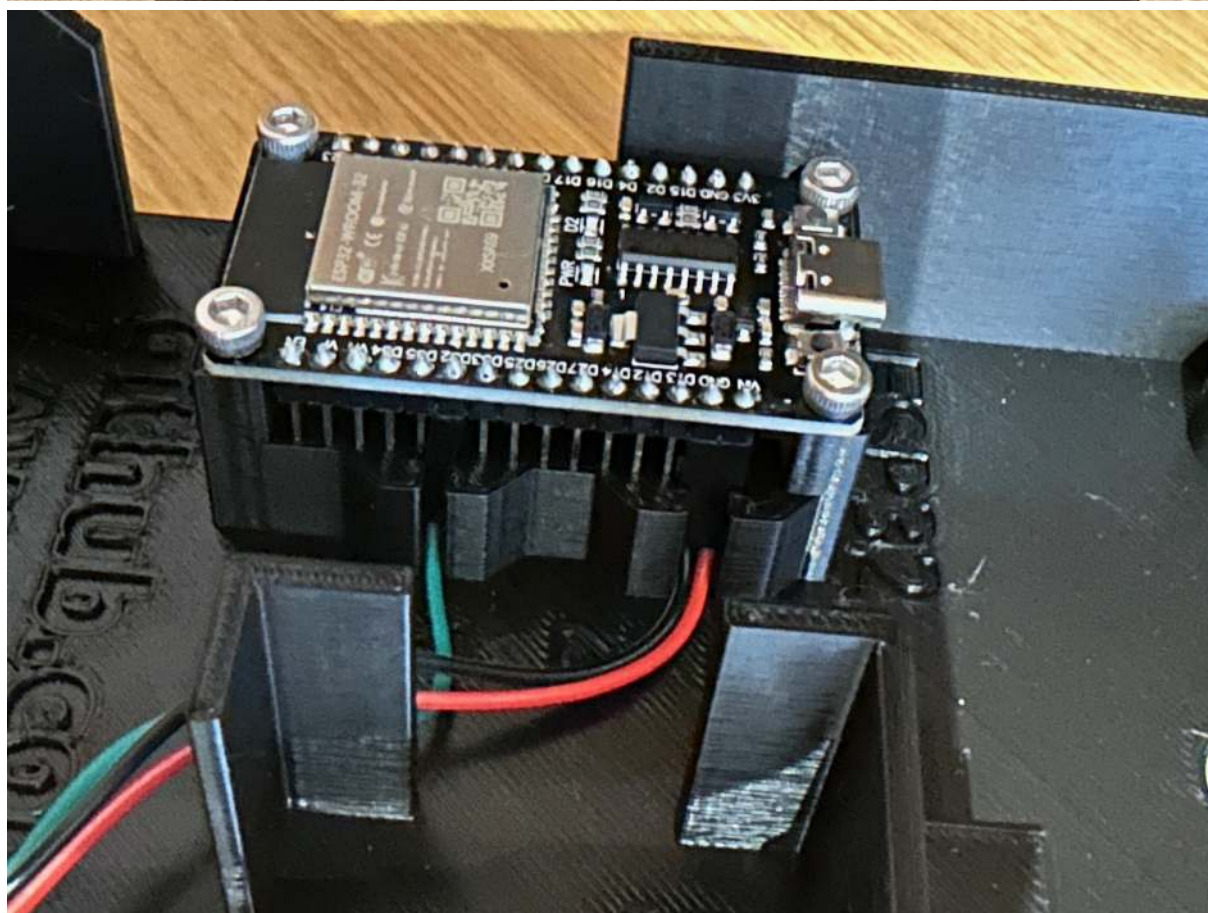
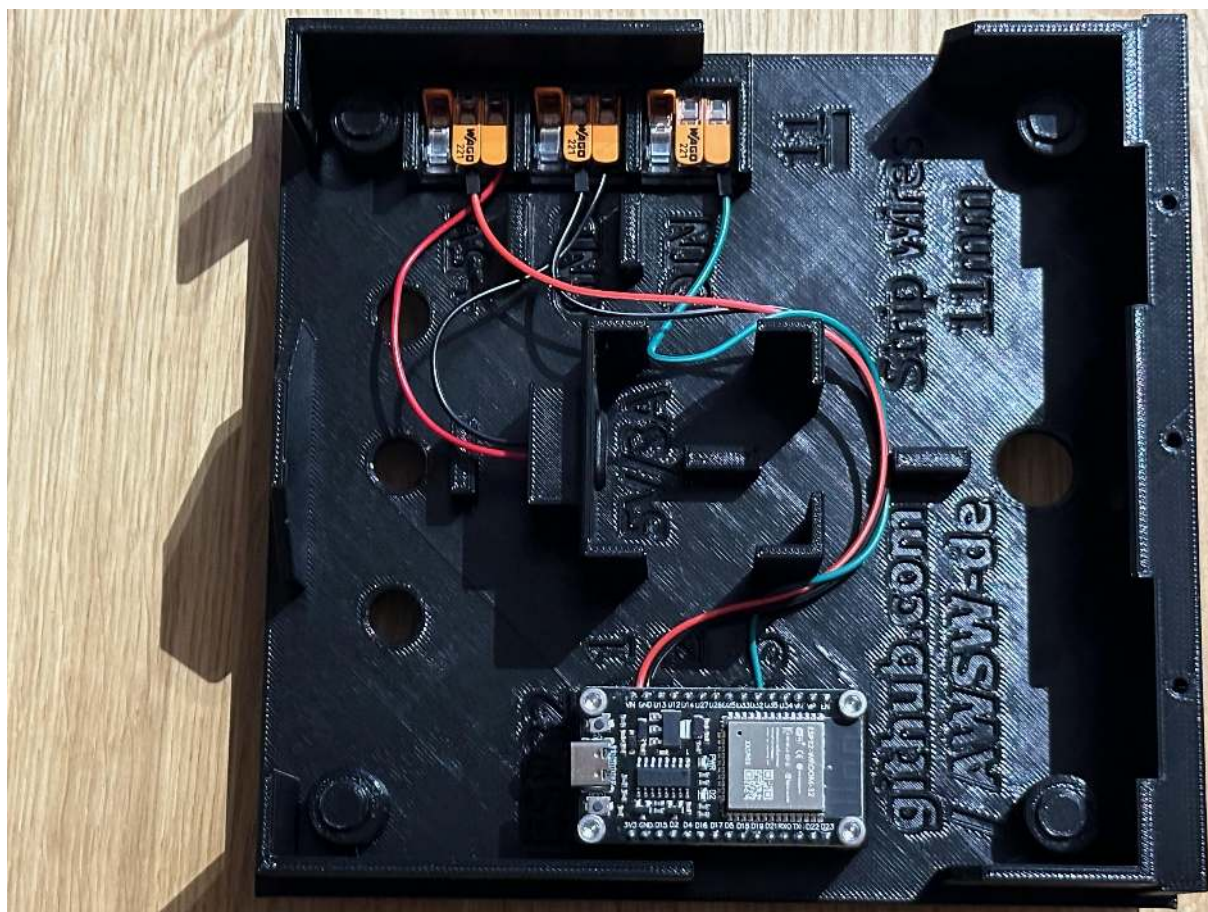
- Add the 2 stripped USB-C port wires to the clamps 1 and 2 as shown and close the clamps on the used slots.



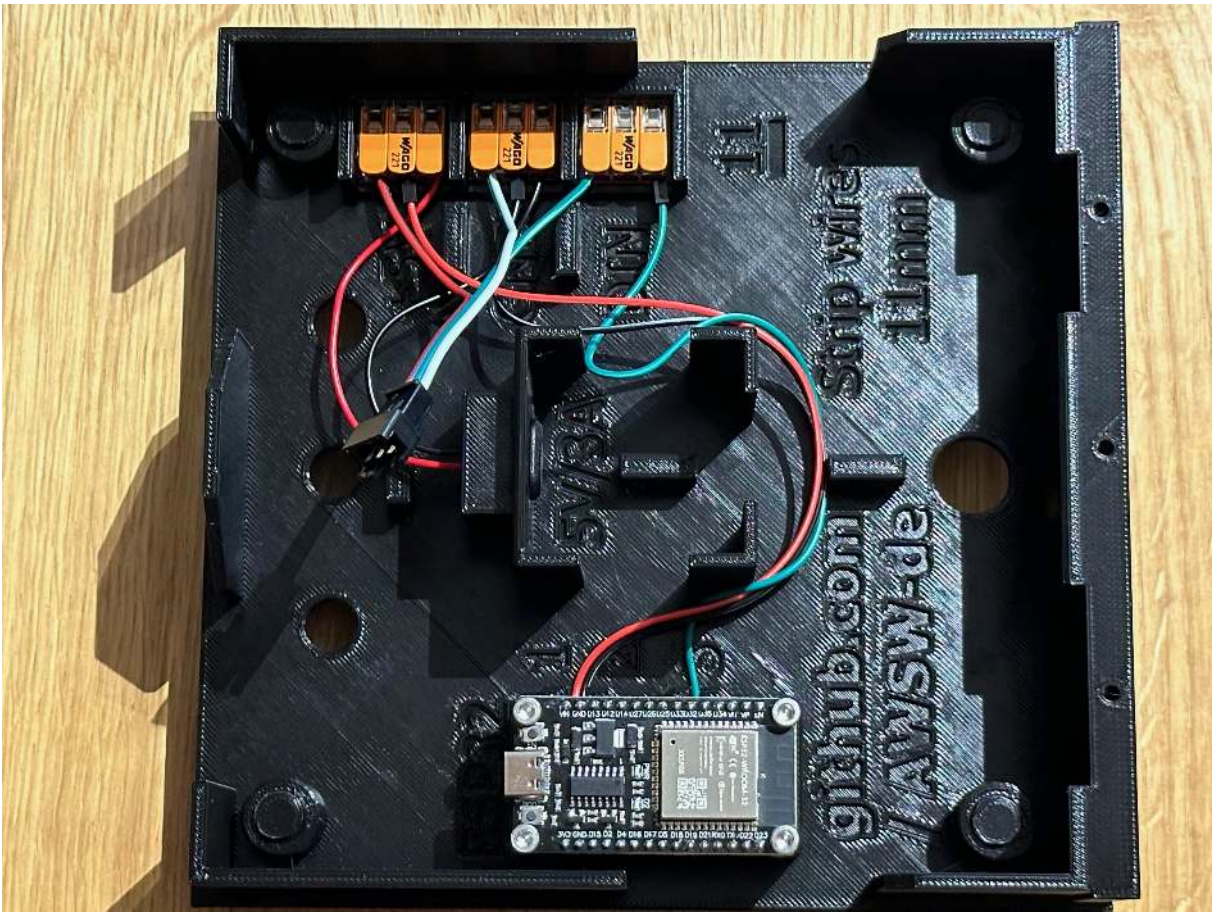
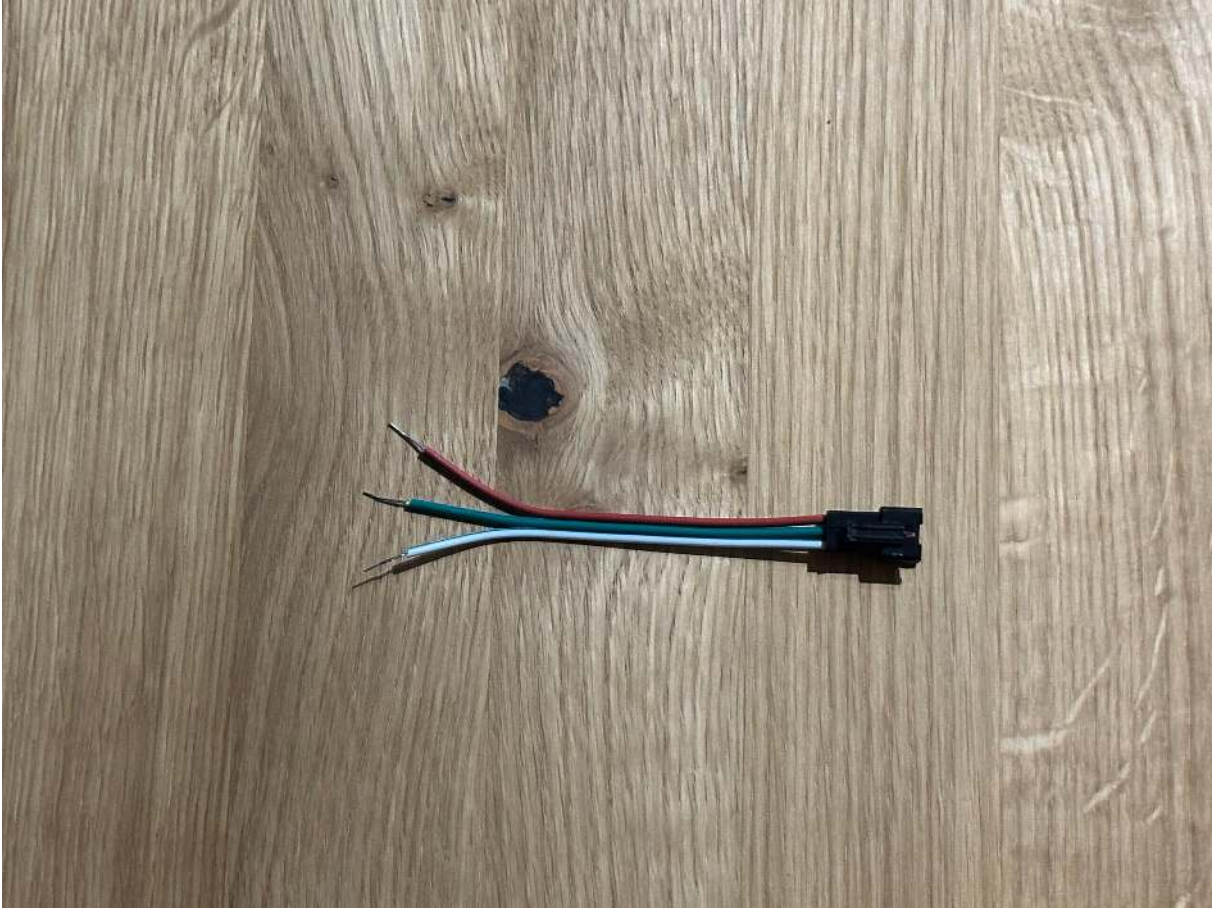
- Add the 3 Female-Male cables like shown to the clamps and close their slots as:
 - 1 = red = 5V
 - 2 = black = GND
 - 3 = green = DIN
- The male pin part of the cable is pushed into the clamps.



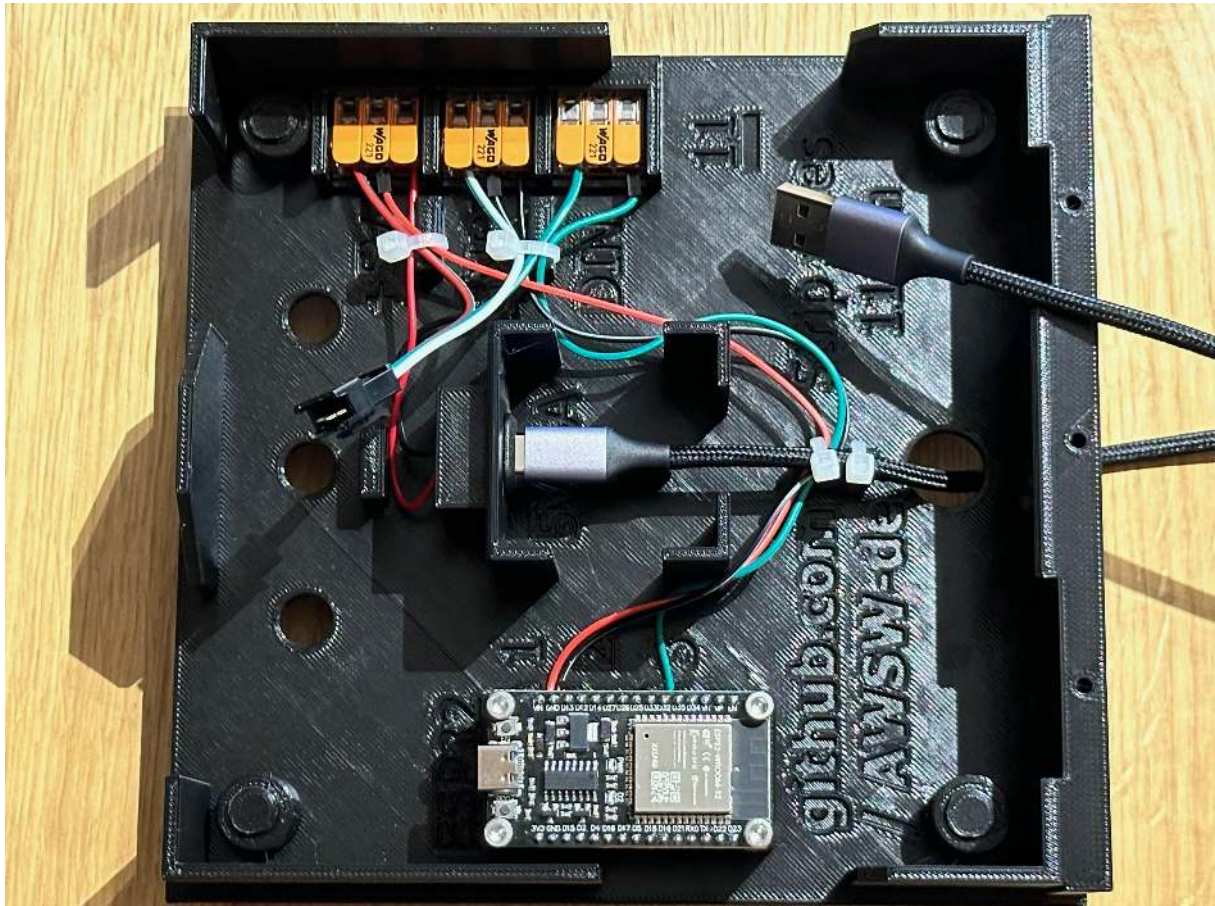
- Add the “female” part of the cables to the ESP32 like shown:
 - VIN = red = 5V
 - GND = black = GND
 - D32 = green = DIN



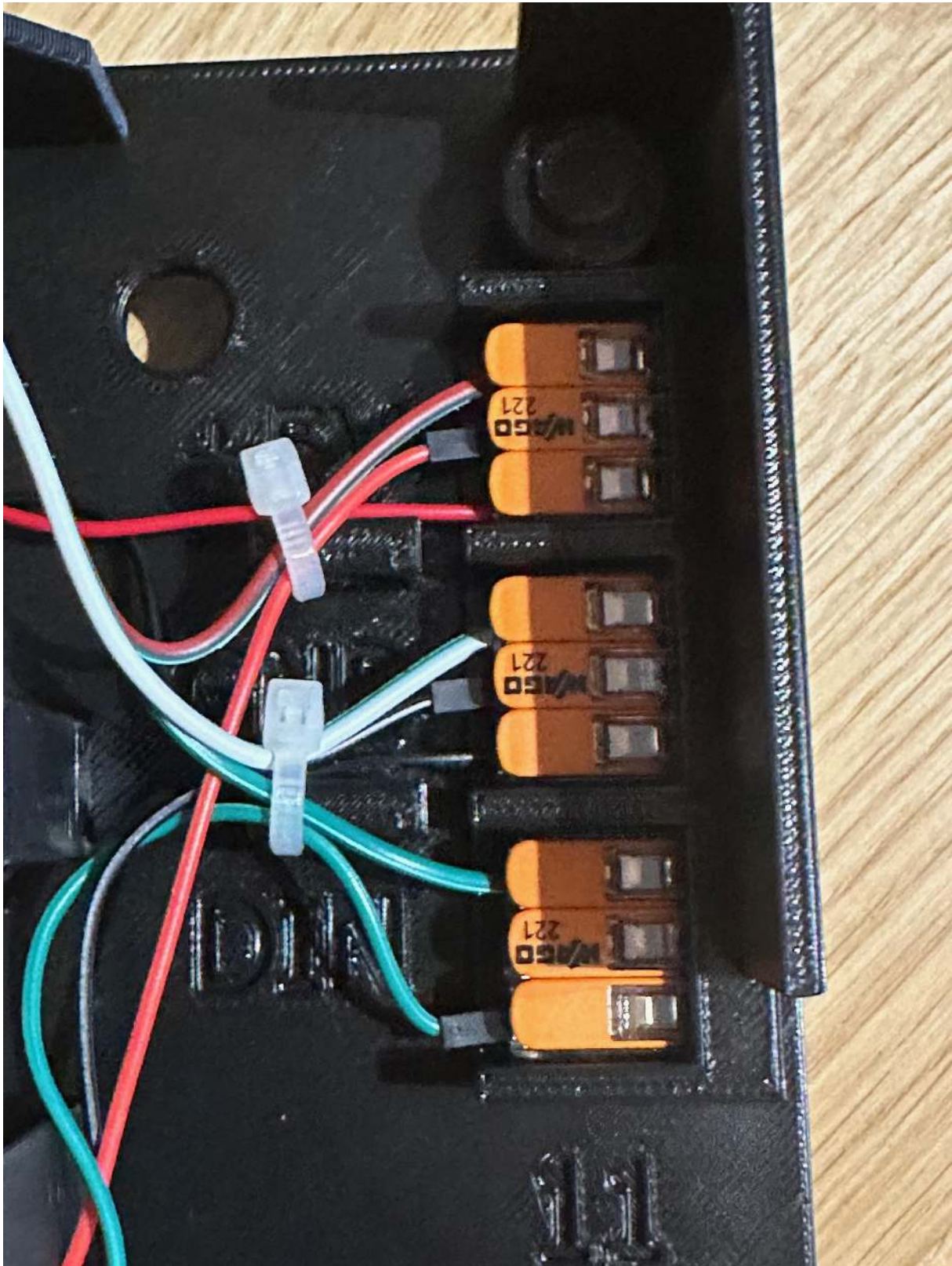
- Carefully push the ESP32 and the 3 cables with their black posts in to the tight holder part of the lid as shown.
- Route the 3 cables out of the small slots.
- Secure the ESP32 in place with the 4 M3x8 screws.



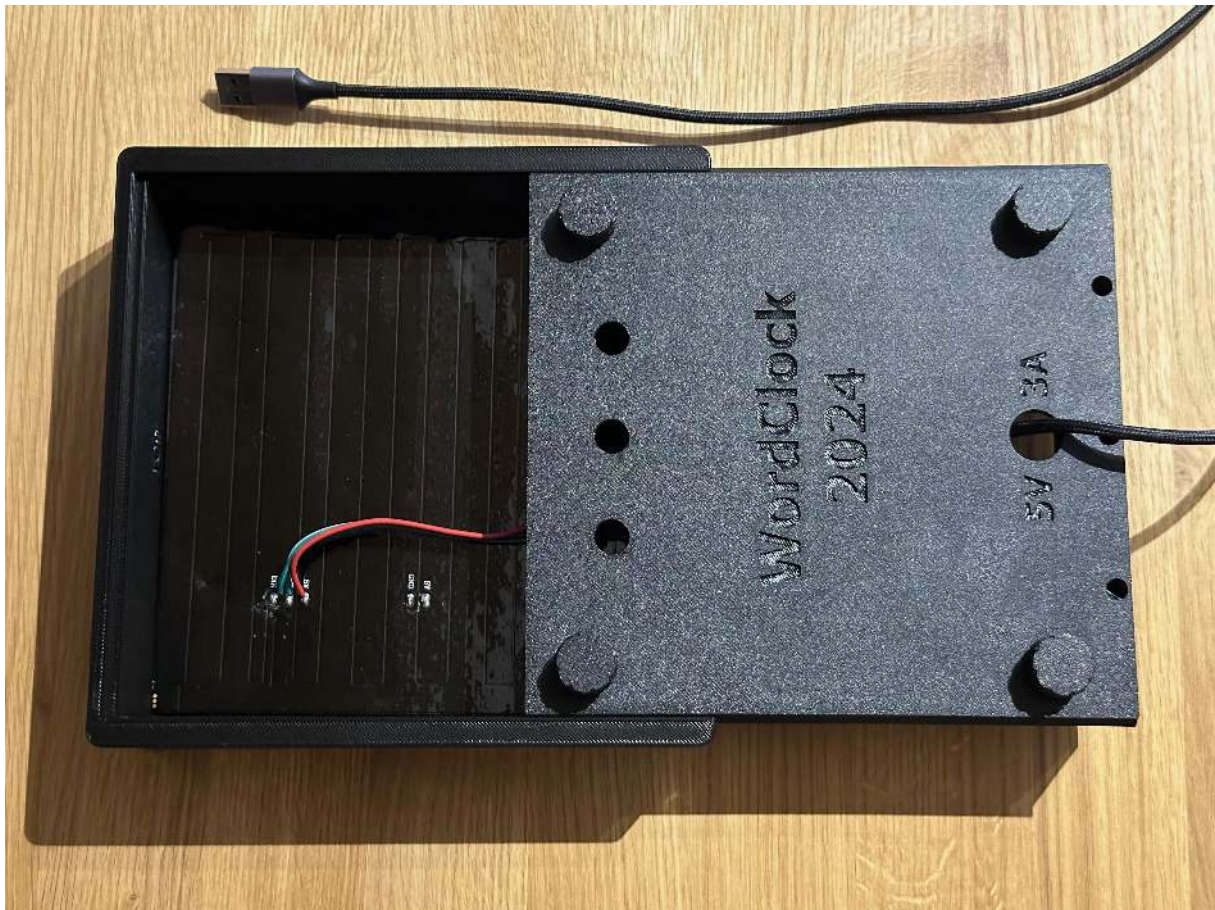
- As last part add the with the LED matrix delivered connector to the circuit.
- Do not forget to strip its wires 11mm too.
- Add the 3 last wires as shown and close the Wago clamps:
 - 1 = red = 5V
 - 2 = white = GND
 - 3 = green = DIN



- Add the USB-C part of the USB-A-to-USB-C cable through its hole in the lid and plug it into the USB-C port like shown.



- If you followed all instructions carefully, checked them again and check that all clamps are closed.
- Please add some small cable ties to secure the cables in place. Especially the power cable should be secured in place.
- If you haven't uploaded the code to the board yet, you should perform this action now!



- As last action you have to do is to connect the LED matrix to its connector in the lid.
- Slide the lid carefully together with the front part as shown.
- The lid pushes the matrix down and holds it in place. No tape or glue etc. needed.
- Use the 3 M3x16 screws to fix the lid in place to the front.
- Plug WordClock to its 5V/3A power supply. Follow the [WiFi setup video](#) for WordClock.
- Check the function and you finished the WordClock setup.
- Enjoy! =)

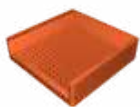
Happy printing =)

Model files

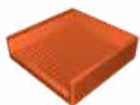


DE - GERMAN

2 files



front-v1-de.3mf

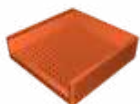


front-v2-de.3mf

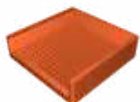


EN - ENGLISH

2 files



front-v1-en.3mf

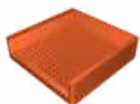


front-v2-en.3mf

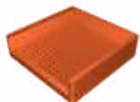


FR - FRENCH

2 files



front-v1-fr.3mf

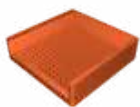


front-v2-fr.3mf

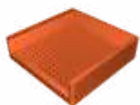


NL - DUTCH

2 files



front-v1-nl.3mf

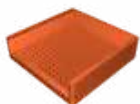


front-v2-nl.3mf

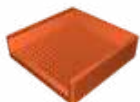


IT - ITALIAN

2 files



front-v1-it.3mf

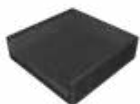


front-v2-it.3mf

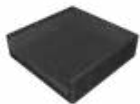


SWE - SWEDISH

2 files



front-v1-swe.3mf

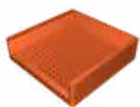


front-v2-swe.3mf

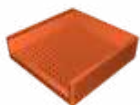


ES - SPANISH

2 files



es-front-v1.3mf

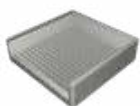


es-front-v2.3mf

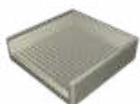


PL - POLISH

2 files



front-v1-pl.3mf

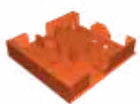


front-v2-pl.3mf



COMMON

5 files



lid.3mf



screw1.3mf



screw2.3mf



screw3.3mf



screw4.3mf

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